# Traffix

(Development)

F24-004-D-Traffix

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# Chapter 1

# Introduction

## 1.1 Problem Statement & Motivation:

With an increasing population, our country has many issues such as increasing traffic that causes delays while traveling. Traffic congestion is one of the biggest challenges Pakistan faces today. The average annual growth rate of Pakistan is 2.55% and Islamabad is 2.80%. [1] Sometimes, it even costs a life when ambulances are not able to reach hospitals on time due to traffic congestion. [2] This shows that an increase in the number of people is the reason for the mismanagement of traffic and transport facilities therefore a stable flow of traffic is very important for enhanced quality of life.

Moreover, there is shortage of traffic police in our country. It has 12,625 employees, and consists of 32 police stations [3]. The current number of traffic police officers does not fulfill the requirement at every signal making it impossible to implement traffic laws and manage traffic at all junctions etc. The ratio of officers to signals is not proportional therefore there is a need for a system that reduces the workload and enhances their work efficiency, allowing them to focus on more critical issues such as dealing with traffic crimes rather than just clearing lanes.

Developed countries have a proper traffic system which contributes to the country's success by reducing traveling time improving road quality and enhancing the lifestyle of people getting them free of everyday struggle of not reaching their workplace on time and to unforeseen traffic congestion. The national highway network is less than 5% of the total road network and caters to about 80% of the commercial traffic. [4] It is very common in Islamabad to face traffic every day, especially early morning when everyone is off to offices and is getting late, so a proper traffic signal system will solve all of these problems and save their time, hence increasing working capacity and productivity.

## 1.2 Problem Solution

We have proposed solutions to the above mentioned problems by introducing our innovative app "Traffix". This will solve many of these problems by smart use of AI and advanced algorithms ultimately conserving energy and resources.

"Traffix" features will have automated traffic signals that will dynamically set the timings and cycle of traffic signals based on car density in that lane or road detected by CCTV footage. This smart system will solve the problem of blocked traffic congestions and hence provide smooth traffic flow without overflow of cars in one certain lane. Consequently, it would reduce overflow of cars in certain lanes improving the travel efficiency.

Furthermore, it will reduce dependency on man force at each signal as signals will manage traffic automatically hence traffic police officers could utilize their time and energy to work on other important matters such as traffic violations and laws. It would reduce mental stress for traffic officers and lead to more contented services.

Moreover, Traffix will be reduced, traffic congestion will cause smooth traffic flow and the emergency vehicles can be detected by CCTV cameras , it will revolutionize the trafficking system. This will allow the emergency vehicles to be detected at a distance and get assisted by traffic signal scheduling to optimize traffic timings such that emergency vehicles are given priority to. It will help save lives by providing clearer routes to ambulances and provide safe traveling. These features should provide improved traffic management and infrastructure for urban environments.

## 1.3 Project Scope

The project "TraffiX" is designed to change traffic management by developing a system that uses real-time CCTV footage, image and video processing with deep learning algorithms. Firstly, object detection using YOLO will real-time identification of various vehicles i.e cars, trucks, motorcycles, and emergency vehicles. Ensuring smooth and smart traffic flow to manage traffic and remove congestion.

Secondly, the system will calculate car density across different lanes, providing data to inform dynamic traffic signal scheduling. This scheduling module will adjust signal timings based on real-time footage of vehicle density, prioritizing lanes with higher traffic to reduce congestion and improve overall traffic flow.

Another challenge of "TraffiX" is the emergency vehicle detection module, which uses various techniques for the training of models to identify ambulances, fire trucks, and other emergency vehicles. The traffic signals will prioritize and adjust the path for the emergency vehicles, making sure these vehicles can navigate through traffic quickly and reach their destinations.

Furthermore, the system includes an automated vehicle challan generation system that identifies traffic violations such as running red lights or stopping beyond designated lines. Using image and video processing, the system captures evidence of these violations and automatically generates challans for the offending vehicles, enhancing traffic rule enforcement and promoting safer driving behavior.

## 1.4 Stake Holders

- Our Team: Zaib Un Nisa, Javaria Habib, Sara Meer Hussain
- Supervisor: Mr. Bilal Khalid Dar
- Co-supervisor: Mr. Wasif Ali Wasif
- FAST-NUCES FYP committee Islamabad
- Traffic Police Officers
- Government

# Chapter 2

# Project Requirement

## 2.1 Use Case Diagram

Here is our Use case diagram for the entire project. It shows how the different external and internal actors interact with the system to ensure a smooth flow of traffic, it contains main actors/users i.e. Traffic police Officer and Admin. It also has two systematic actors i.e. CCTV camera and Smart Traffic Controller. It also has **<<include>>** which means that if there is a traffic violation it is mandatory for challan generation and on the other hand, **<<extend>>** which allows the user to have the option to either choose manual or automatic setting.

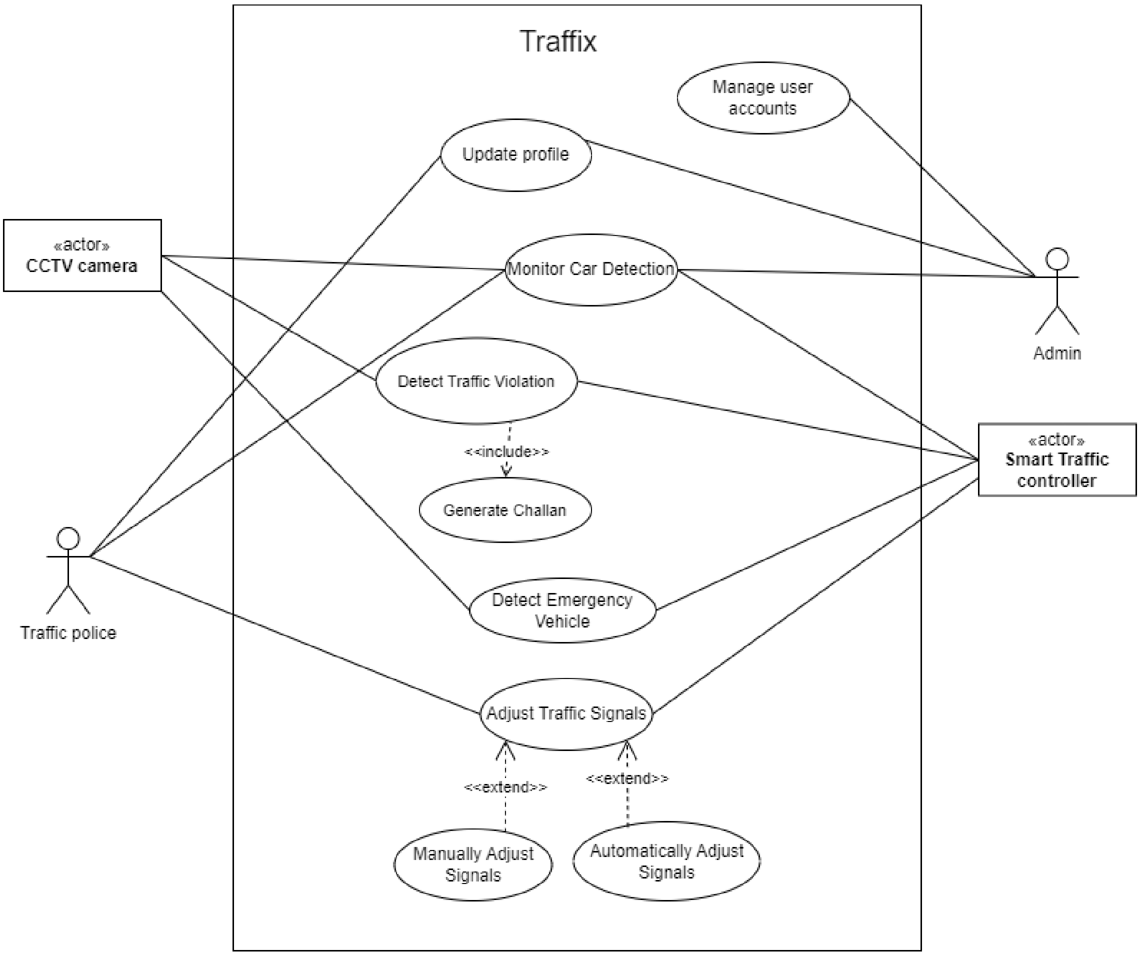



*Figure 1: Use Case Diagram*

## Brief Use Cases:

- Update Profile

| Use case | USC-1 Update Profile |
|---|---|
| Actor | User |
| Type | Primary |
| Description | The user can update their personal information such as contact number , password. |

*Table 1: USC-1 Update Profile*

- Monitor Car Detection

| Use case | USC-2 Monitor Car Detection |
|---|---|
| Actor | Traffic Police, Admin |
| Type | Primary |
| Description | Traffic police and the admin can view real-time CCTV footage to monitor traffic which will later include it in the dynamic system. |

*Table 2: USC-2 Monitor Car Detection*

- Detect Emergency Vehicle

| Use case | USC-3 Detect Emergency Vehicle |
|---|---|
| Actor | Smart Traffic Signal System |
| Type | Automatic |
| Description | The system will use video processing to detect the incoming emergency vehicles. |

*Table 3: USC-3 Detect Emergency Vehicle*

## Fully dressed Use cases:

- Update Profile

| Use case Name | USC-1 Update Profile |
|---|---|
| Scope | Traffix |
| Level | User Goal |
| Primary Actor | User |

| | |
|---|---|
| **Stakeholders & Interests** | User: Wants to keep their personal info up to date. |
| **Precondition** | The user is assigned a profile and is logged into the system. |
| **Success Guarantee** | The updated information is saved in the database. |
| **Main Success Scenario** | 1. User selects the “Update profile” option from side bar.<br>2. The system displays the current profile info.<br>3. The user makes any new changes.<br>4. The user clicks on the save button.<br>5. The system validates the information.<br>6. The system saved the new information successfully in the database.<br>7. The system prompts the confirmation message. |
| **Extensions** | • 5a. If the validation fails (i.e incorrect phone number format, short length of new password), then the system prompts to update the information again. |
| **Special Requirements** | N/A |
| **Technology & Data Variance List** | N/A |
| **Frequency of occurrence** | Depends on user – Periodically |
| **Miscellaneous** | N/A |

*Table 4: Fully dressed USC-1*

- Monitor Car Detection

| **Use case Name** | **USC-2 Monitor Car Detection** |
|---|---|
| **Scope** | Traffix |
| **Level** | User Goal |
| **Primary Actor** | Traffic Police, Admin |
| **Stakeholders & Interests** | Traffic police: Wants to monitor traffic via real-time CCTV footage of vehicles |
| **Precondition** | Functional CCTV cameras<br>The user is logged into the system to view footage. |

| | |
|---|---|
| **Success Guarantee** | Real-time traffic is monitored by the officers and admin. |
| **Main Success Scenario** | 1. The actor has accessed the system.<br>2. The system displays CCTV footage of movement of vehicles.<br>3. The actor observes the footage.<br>4. The actor takes any necessary actions. |
| **Extensions** | • 2a. If the real-time footage isn't available, then the actor fixes it first. |
| **Special Requirements** | The system should let the streaming of footage with low latency. |
| **Technology & Data Variance List** | Internet connection may affect the receiving of real-time footage. |
| **Frequency of occurrence** | Continuous. |
| **Miscellaneous** | N/A |

*Table 5:Fully dressed USC-2*

- Detect Emergency Vehicle

| **Use case Name** | **USC-3 Detect Emergency Vehicle** |
|---|---|
| **Scope** | Traffix |
| **Level** | Subsystem |
| **Primary Actor** | Smart Traffic Signal System |
| **Stakeholders & Interests** | System: Wants to ensure the emergency vehicles are detected separately for priority passage. |
| **Precondition** | The smart traffic system is working with CCTV cameras for real-time footage.. |
| **Success Guarantee** | Emergency vehicles are identified. |
| **Main Success Scenario** | 1. The system monitors traffic via CCTV cameras.<br>2. The system identifies emergency vehicles using video processing.<br>3. The system automatically adjusts the traffic signal to prioritize the passage of emergency vehicle.<br>4. The system alerts the traffic police of emergency vehicle. |

| | |
|---|---|
| **Extensions** | • 2a The system remains idle until an emergency vehicle is detected. |
| **Special Requirements** | N/A |
| **Technology & Data Variance List** | CCTV camera' type might affect the accuracy of detection. |
| **Frequency of occurrence** | Based on incoming emergency vehicles. |
| **Miscellaneous** | N/A |

*Table 6:Fully dressed USC-3*

## 2.2 Modules

We have six different modules in our project, which are as follows:

### 2.2.1 Traffic Detection from CCTV

In this module, the CCTV camera will monitor traffic and the footage will be integrated with image and video detection algorithms to control and manage traffic.

### 2.2.2 Traffic Signal Scheduling

In this module, the system will use the videos detected by CCTV cameras and implement a scheduling algorithm on them to dynamically control traffic. The opening/ closing of the signal will be managed by the designed algorithm.

### 2.2.3 Emergency Vehicle Detection

In this module, the emergency vehicle detection module will detect the vehicles like ambulances, fire trucks, and police cars using video detection and then modify the traffic signal timings to provide a clear path for the emergency vehicle.

### 2.2.4 Automated vehicle challan generation system

In this module, the vehicles will be issued a challan for violating the traffic rules i.e breaking a red light.

### 2.2.5 Manual setting of signals

In this module, the traffic police will be able to manually set the time of the signals in case of any emergency or uncertainty, i.e. VIP protocol, protests, etc.

### 2.2.6 Admin Dashboard

In this module, the admin will be able to manage settings, monitor traffic, and manage profiles. The admin also ensures the overall working of the system.

## 2.3 User Classes and Characteristics

| User Class | Description |
|---|---|
| **Traffic Police Officer** | The traffic police officer uses the system integrated at their specified location. The officer ensures the system is working and integrated properly. They are also able to monitor traffic/change signal settings. They can view the challans generated and emergency vehicle alerts. |
| **Admin Officer** | The admin officer manages the traffic police officers and their locations. The admin can register new users and assign them a location. The admin officer can monitor/change signal settings. |

*Table 7: User Classes & Characteristics*

# 2.4 Functional Requirements

## 2.3.1 Detection of vehicles

- The system shall distinguish between different types of vehicles.
- The system shall detect moving vehicles.
- The system shall detect stationary vehicles.

## 2.3.2 Calculation of Car Density

- The system shall calculate the total number of vehicles.

## 2.3.3 Traffic Signal Scheduling

- The system shall prioritize lanes with more vehicle density.
- The system shall adjust traffic signal timings based on real- footage of vehicle density.
- The system shall adapt to changing traffic conditions throughout the day.

## 2.3.4 Emergency Vehicle Detection

- The system shall detect emergency vehicles in traffic using image detection.
- The system shall modify traffic signal timings to provide a clear path for detected emergency vehicles.
- The system shall notify traffic control when an emergency vehicle is detected.

### 2.3.5 Automated vehicle challan detection system

- The system shall identify traffic violations i.e. breaking the signal etc.
- The system shall record and store data on detected violations.

### 2.3.5 Admin Dashboard

- The admin shall be able to register new Traffic Officers.
- The admin shall be able to view the Traffic Officers.
- The admin shall be able to update the Traffic Officer’s location.
- The admin shall be able to delete the Traffic Officer’s accounts.
- The admin shall be able to view the Traffic Officers.
- The admin shall be able to monitor Traffic.
- The admin shall be able to manage Signal Control.

## 2.5 Non-Functional Requirements

### 2.1.1 Reliability

RT-1: The system shall have a 24-48 hrs Mean Time Between Failures (MTFB), ensuring continuous operation except for scheduled maintenance.

### 2.1.2 Usability

USE-1: The system shall have a user-friendly interface, allowing Traffic Police to use the system efficiently.

### 2.1.3 Performance

PR-1 The system shall maintain its average response time of less than 5 seconds in various conditions etc.

### 2.1.4 Security requirements

SEC-1: The system shall require user authentication for different tasks depending on user's role.

## 2.6 Domain Model

The domain model diagram of the system “Traffix” comprehensively shows the key entities and their relationship with each other. It also shows the entire system works in an integrated manner in Traffix System with its internal and external entities working to manage the system and its users.

Here is the diagram as follows:

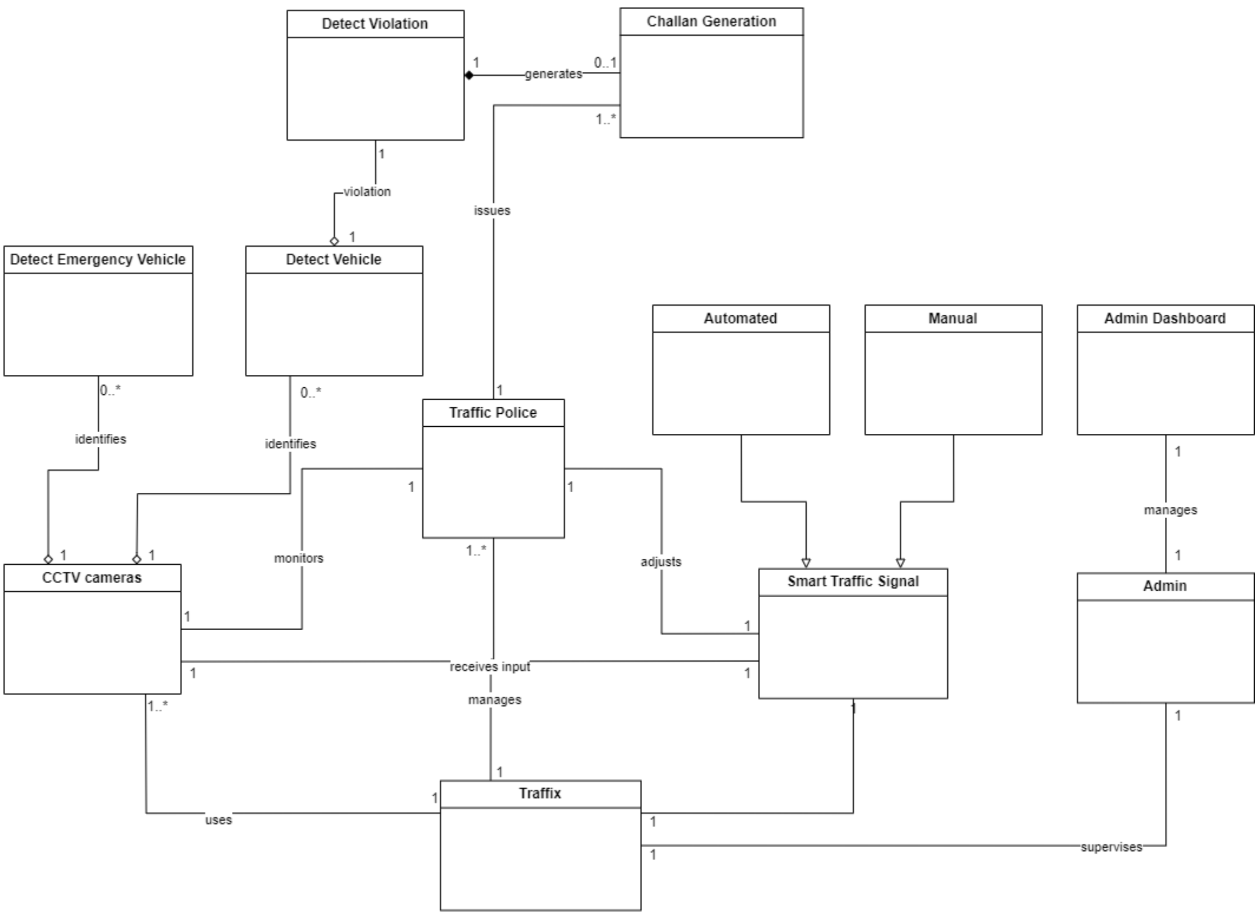



*Figure 2: Domain Model Diagram*

# Chapter 3

# System Overview

## 3.1 Architectural Design

The diagram that best fits our system is the Event Driven Architecture since our system enables our system “Traffix” to detect the events occurring and act on them in real-time. As our system relies upon real-time CCTV footage to detect traffic flow any event such as emergency vehicle detection, traffic violation detection, or car detection can generate instant automated responses for them. Our system comprises of complex systems to cover large areas with input from CCTV footage. It helps handle our system’s complex and time-sensitive tasks efficiently.

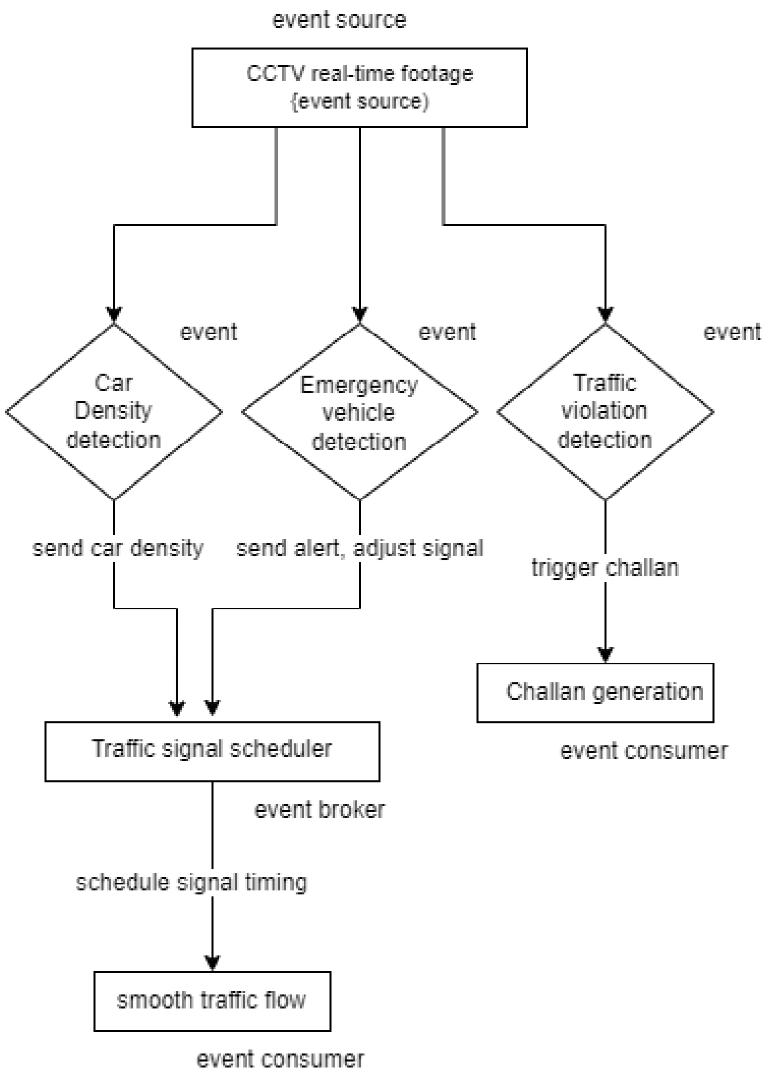



*Figure 3: Architecture Diagram*

## 3.2 Design Models

We have chosen procedural approach design models rather than OOP methodology due to the requirements of our system, “Traffix”. Since it is comprised of data processing and step-by-step procedural logic for processing data and generating responses based on any event that occurs. The flow of diagrams will explain the complexity of our system and how the process of each event works.

We are using the following design models:

- Activity Diagram
- Data Flow Diagram
- State Transition Diagram
- System Level Sequence Diagram (SSD)
- Sequence Diagram (SD)

## 3.2.1 Activity Diagram

The following are the activity diagrams for our system.

Update Profile Activity Diagram

The following is the activity diagram for update profile

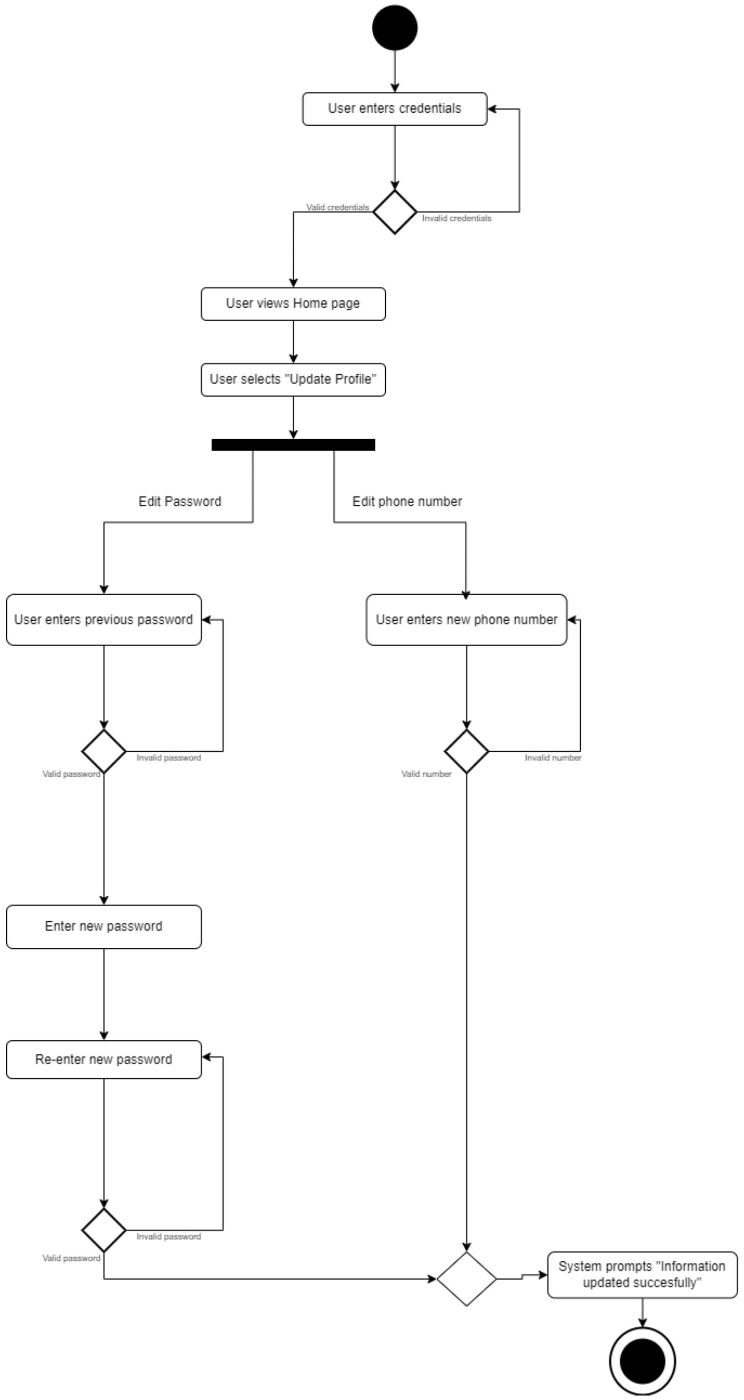



*Figure 4: Update Profile - Activity Diagram*

## Monitor Car Detection Activity Diagram

The following is the activity diagram for Monitor Car Detection.

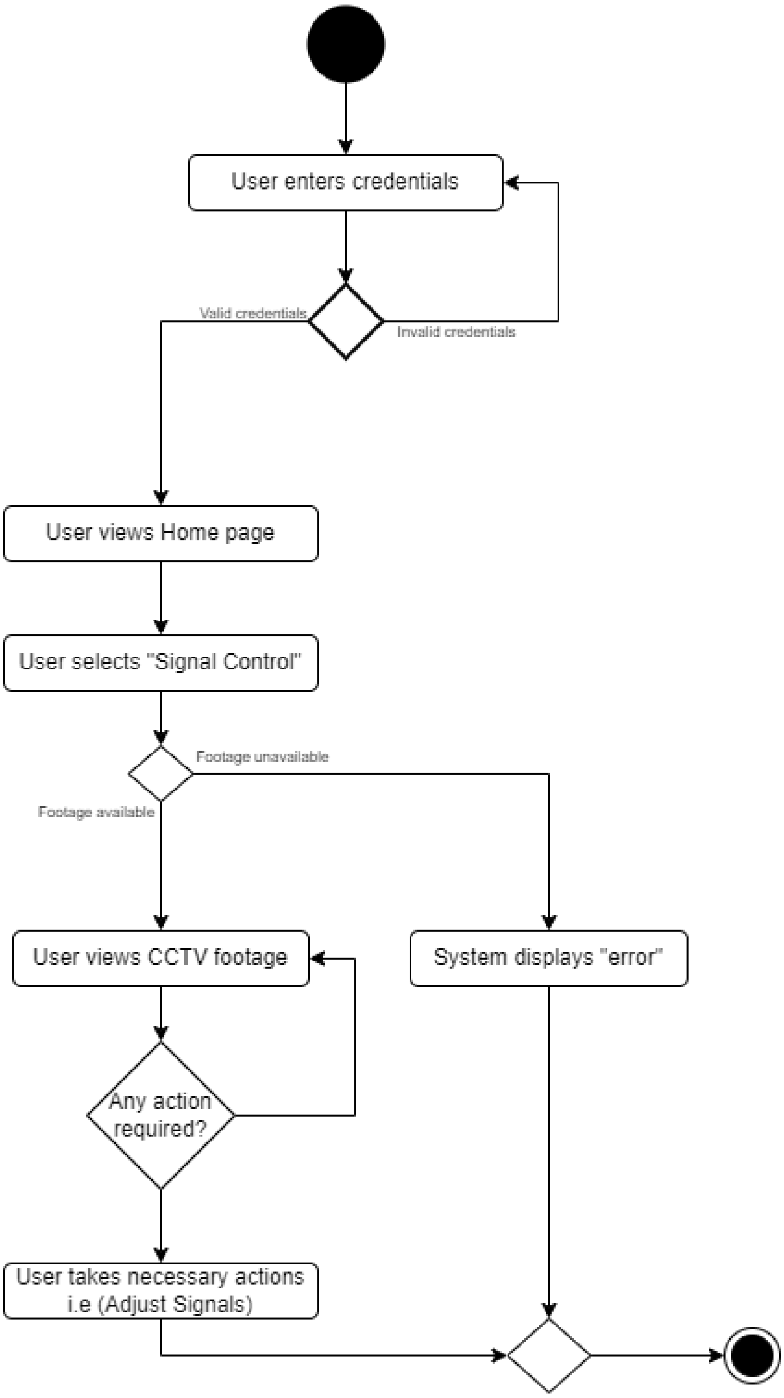



*Figure 5: Monitor Car Detection - Activity Diagram*

## Detect Emergency Vehicle Activity Diagram

The following is the activity diagram for Detect Emergency Vehicle

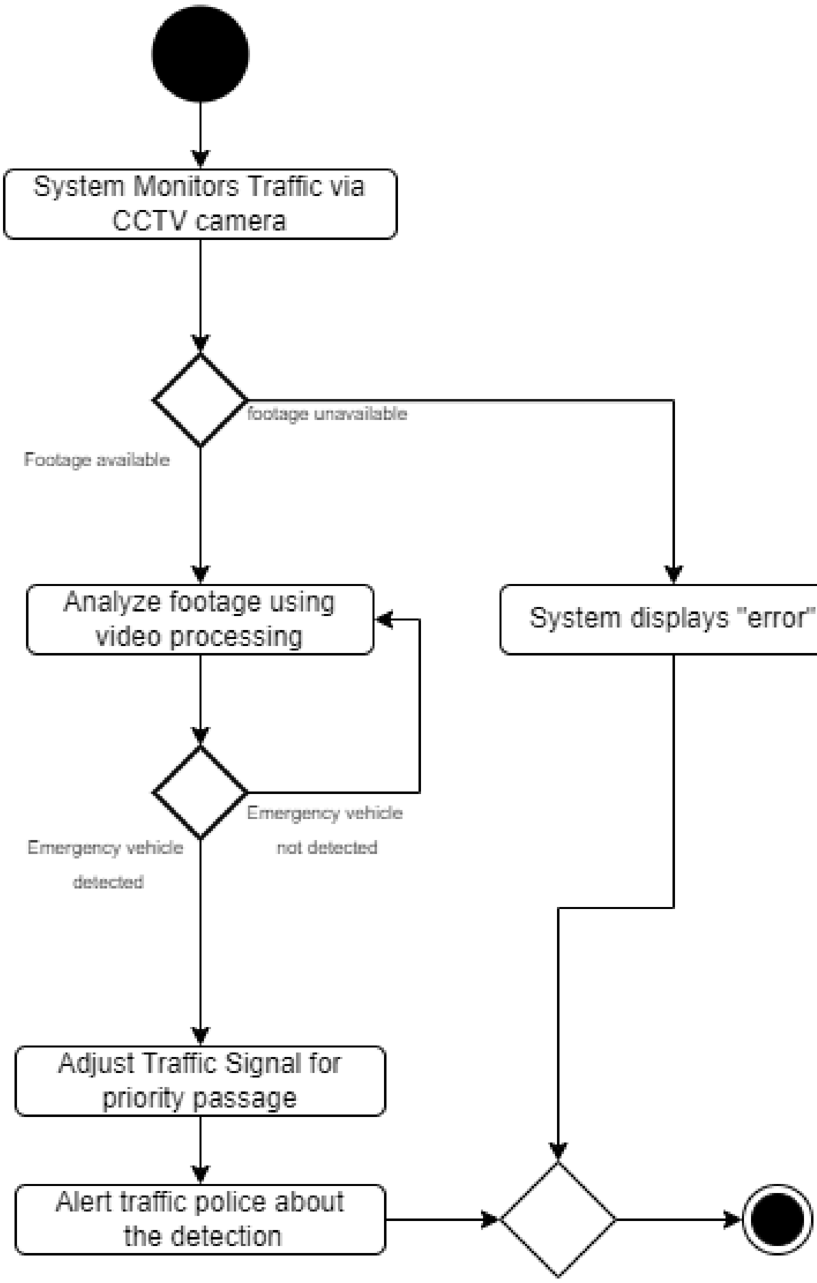



*Figure 6: Detect Emergency Vehicle - Activity Diagram*

## 3.2.2 Data flow Diagram

For graphical representation of the data flow within Traffix we have made use of a modelling diagram that provided a dynamic view of the system i.e  Data Flow Diagram (DFD)

### DFD Level 0

This is the context level, in which we are giving an abstract overview of the system.

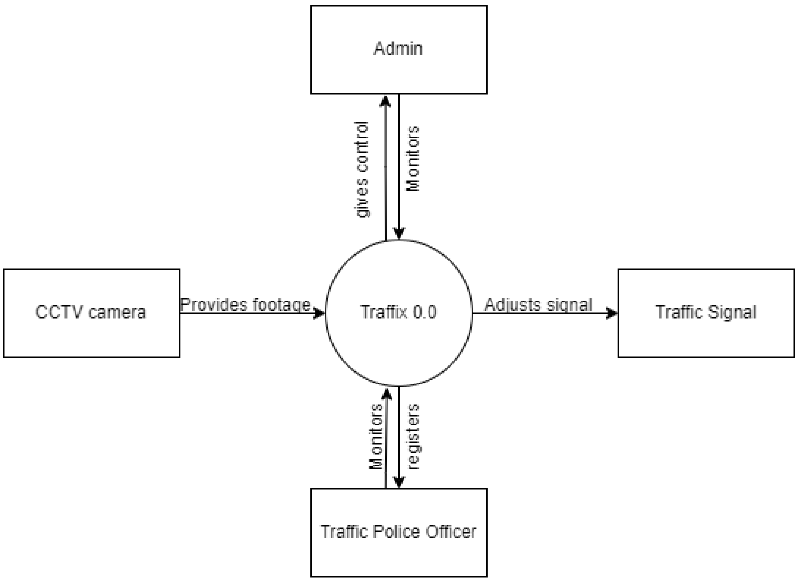



*Figure 7: DFD Level 0*

### DFD Level 1

The is the Level 1, a more detailed view of the system where it shows all the processes involved in the entire processing of dynamically adjusting traffic.

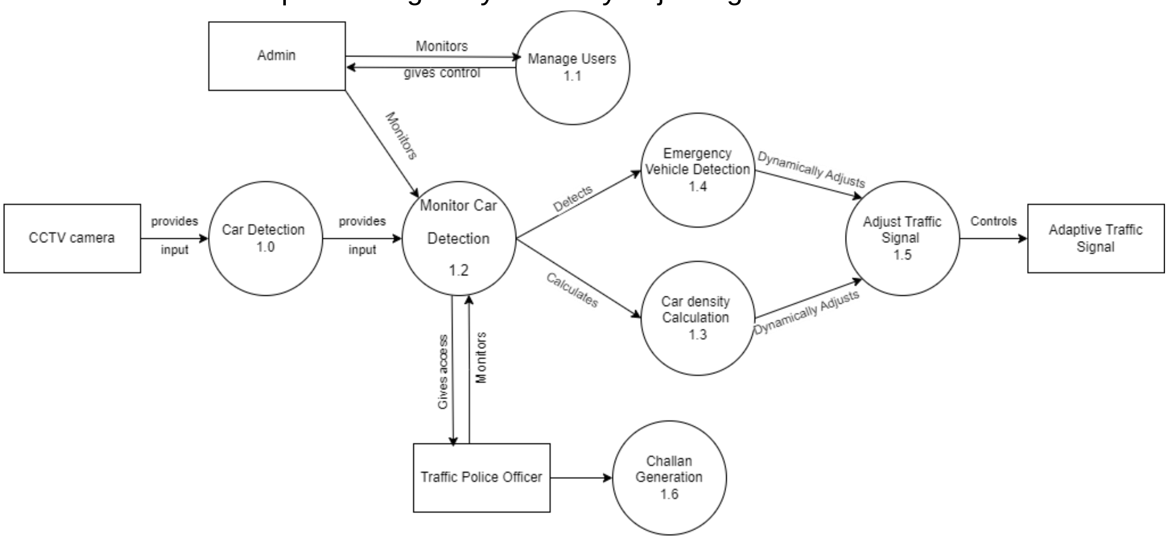



*Figure 8: DFD Level 1*

## DFD Level 2

In this Level 2, it shows the database of footage detection and database for user data with the entire processes and its sub-processes.

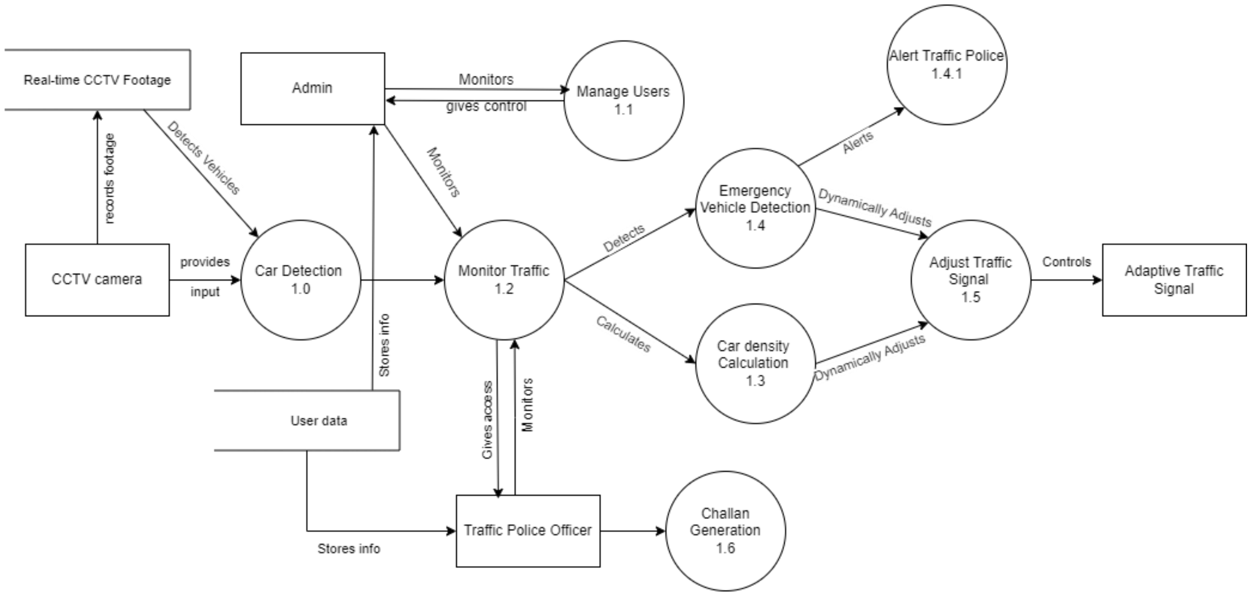



*Figure 9: DFD Level 2*

### 3.2.3 System-level Sequence Diagram (SSD)

To model operations initiated between the users and system, Traffix, we have made use of system sequence diagrams.

#### Update Profile

The following is the ssd for update profile, for updating password and phone number.

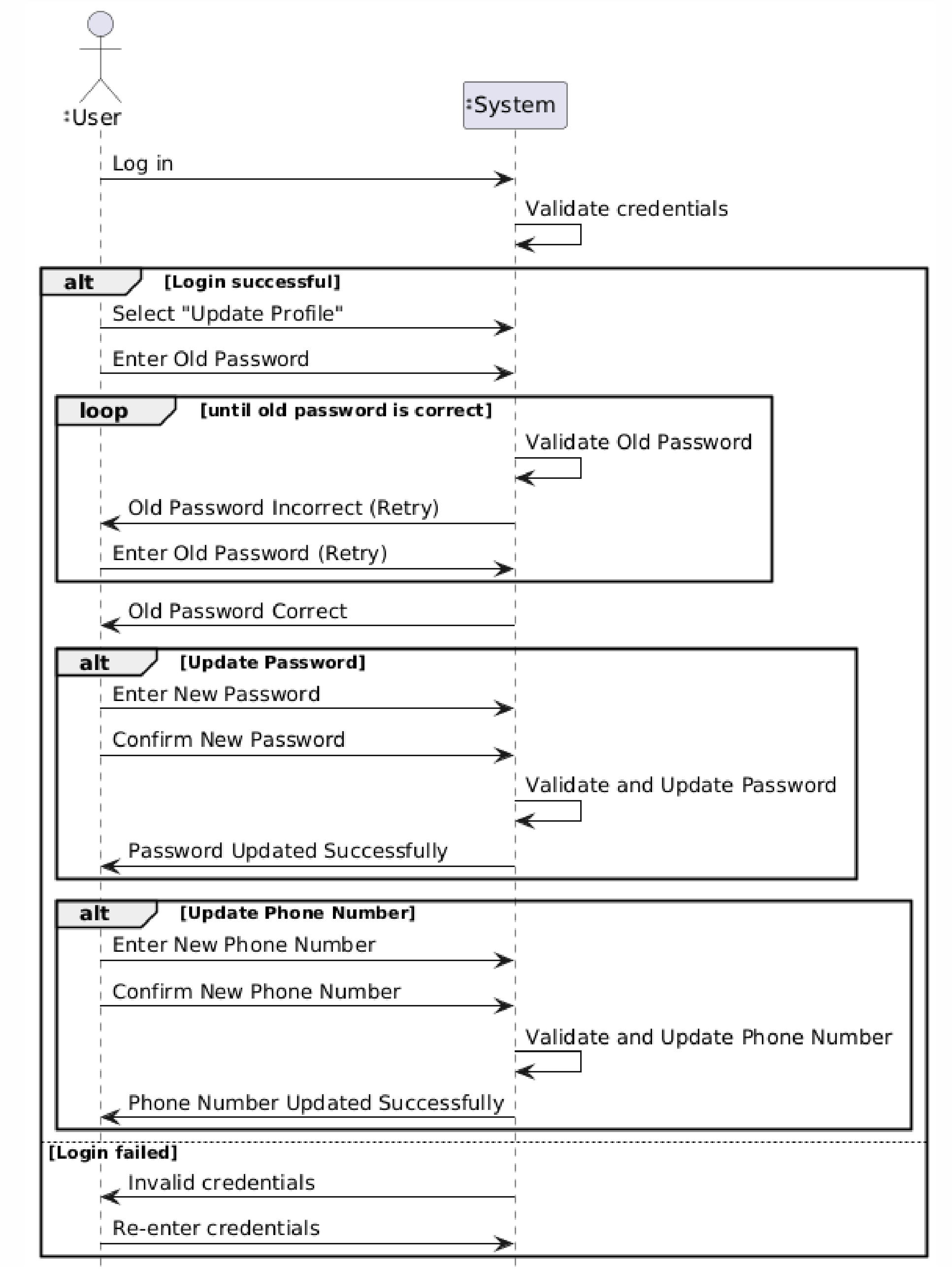



*Figure 10: Update Profile SSD*

## Monitor Car Detection

The following is the ssd for monitor car detection.

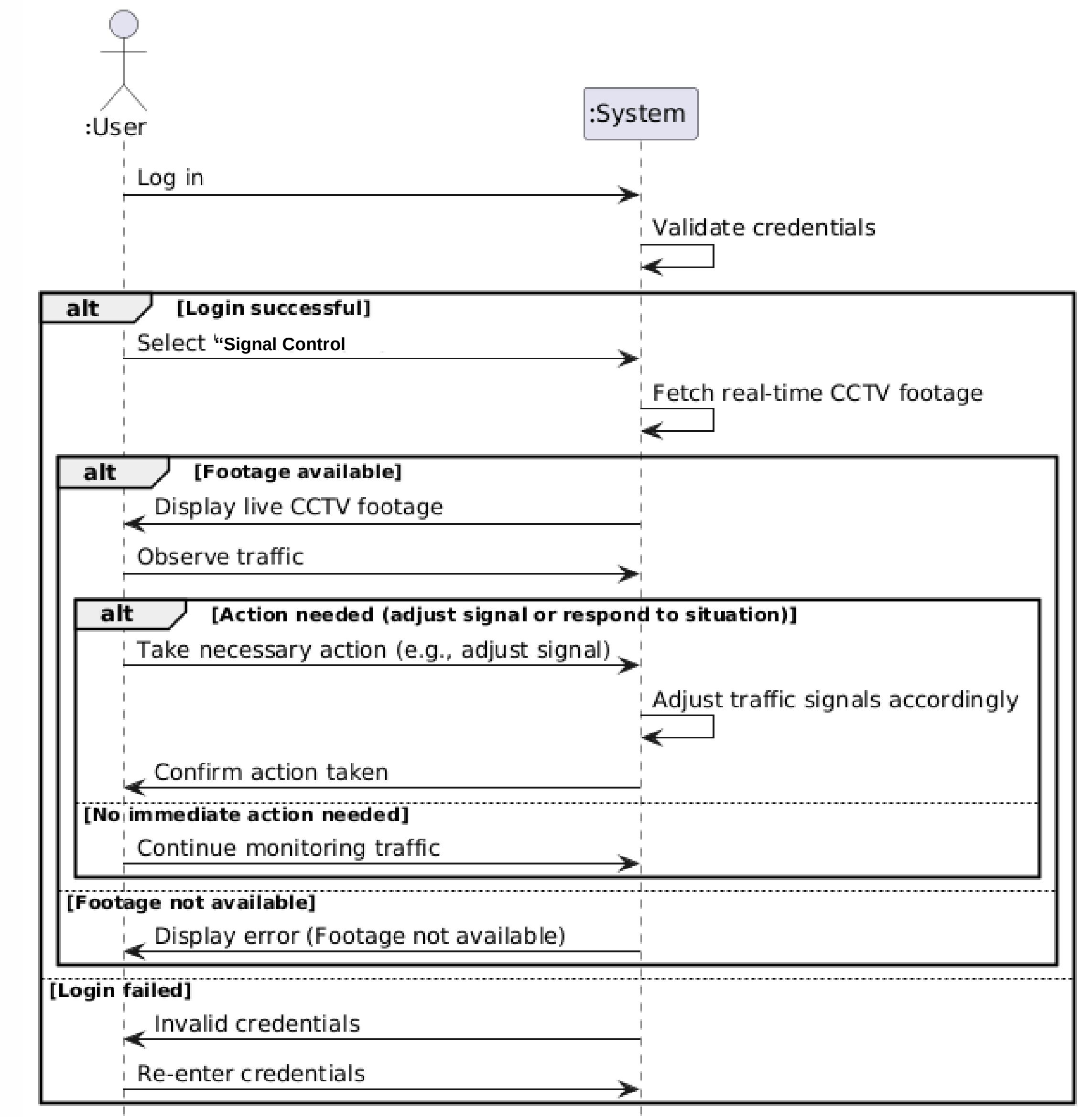



*Figure 11: Monitor Car Detection SSD*

## Detect Emergency Vehicle

The following is the ssd for detection of emergency vehicle detection.

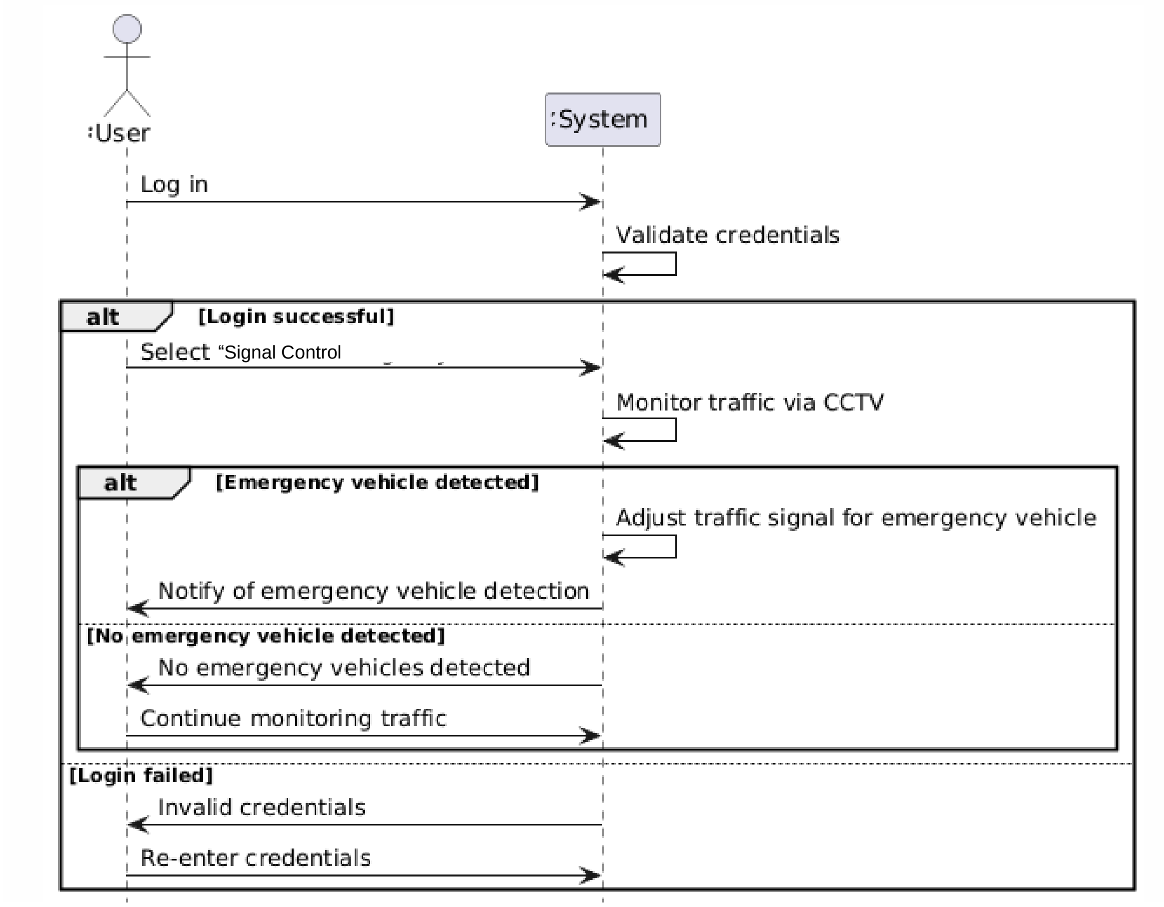



*Figure 12: Emergency vehicle detection SSD*

## 3.2.4 Sequence Diagrams (SD)

To model operations initiated between the users and the sub systems working all together in Traffix, we have made use of system sequence diagrams.

### Update Profile

The following is the sequence diagram for update profile, for updating password and phone number.

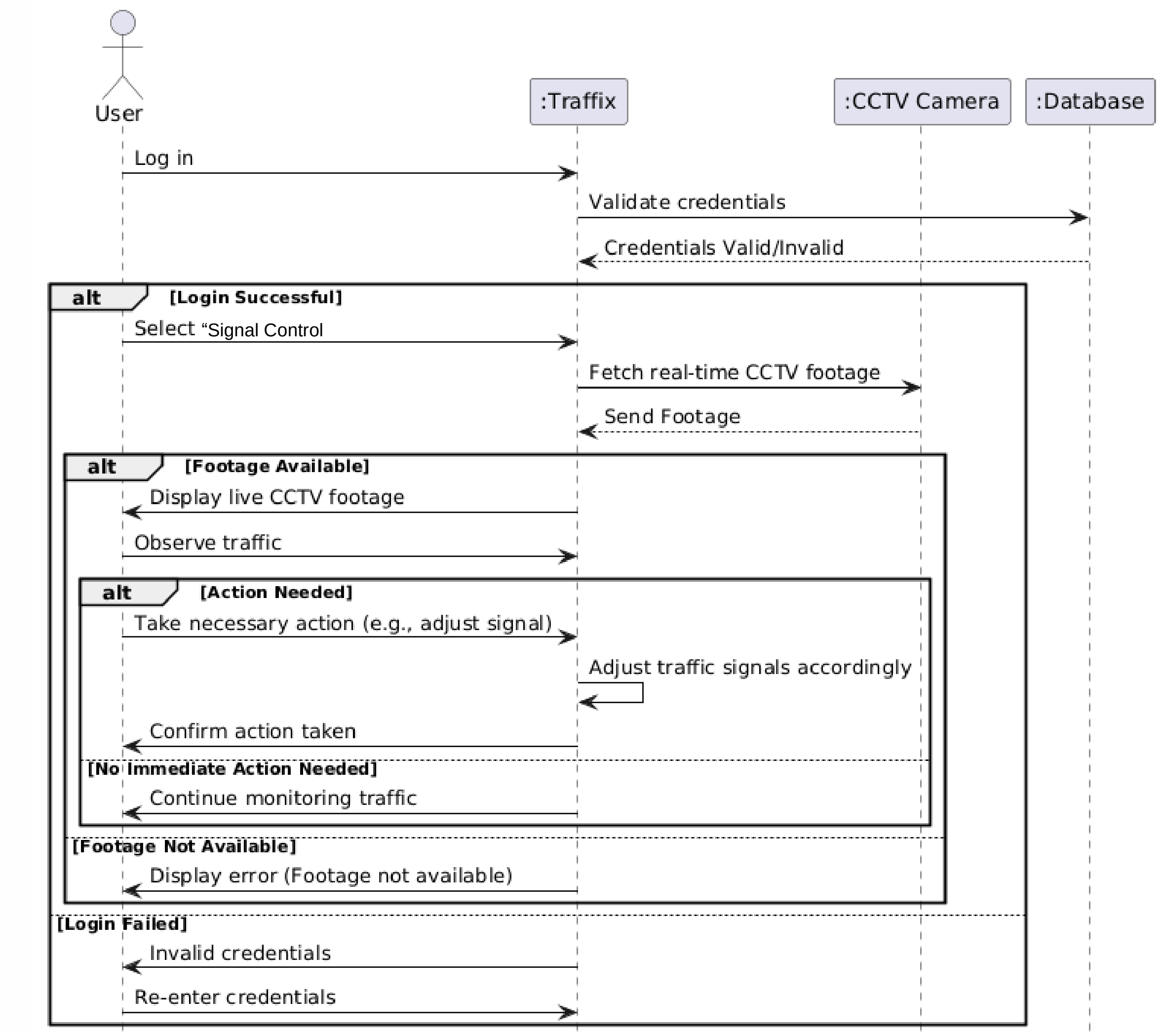



*Figure 13: Update Profile SD*

## Monitor Car Detection

The following is the sequence diagram for monitor car detection.

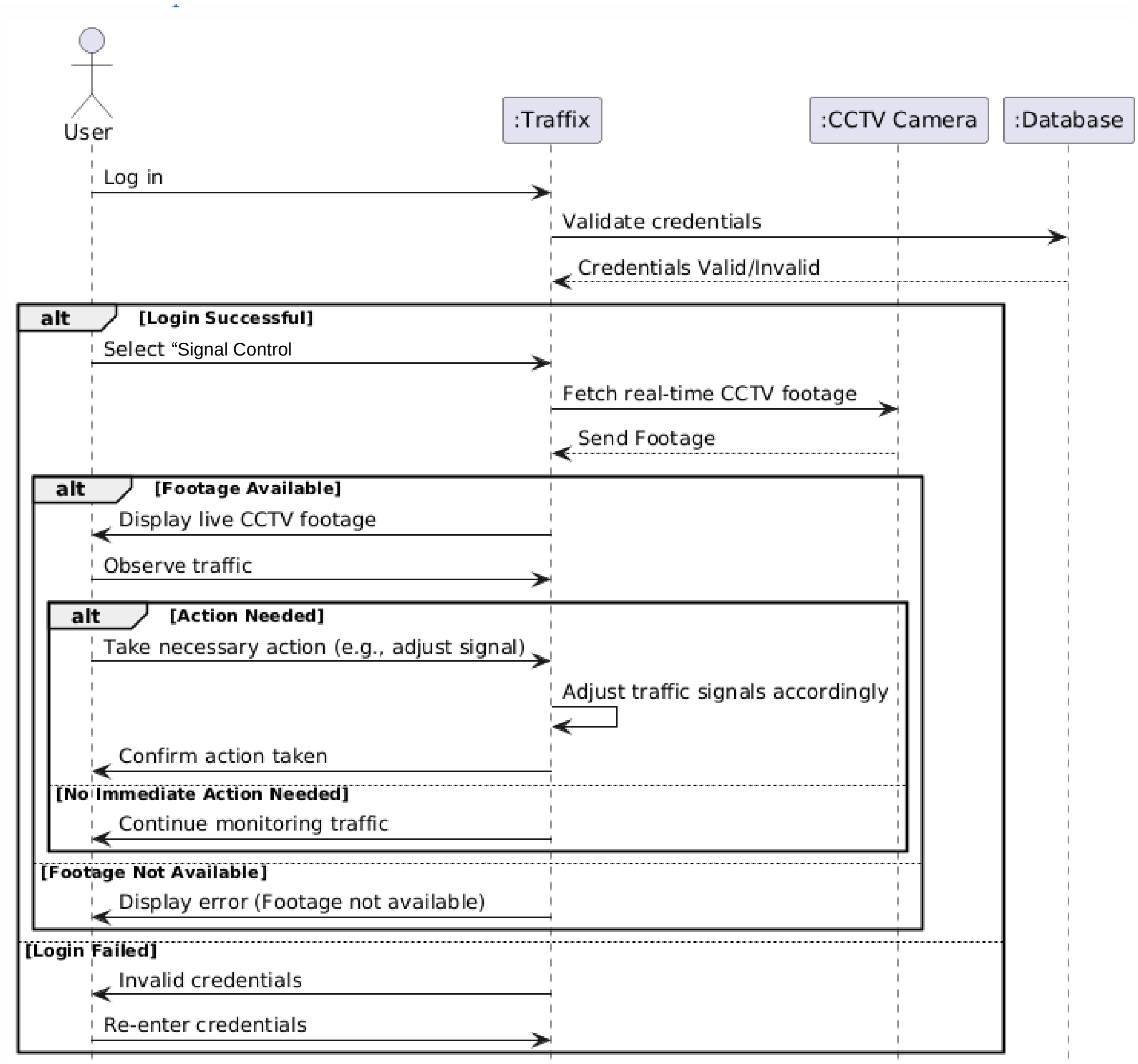



*Figure 14: Monitor Car Detection SD*

## Detect Emergency Vehicle

The following is the sequence diagram for detection of emergency vehicle detection.

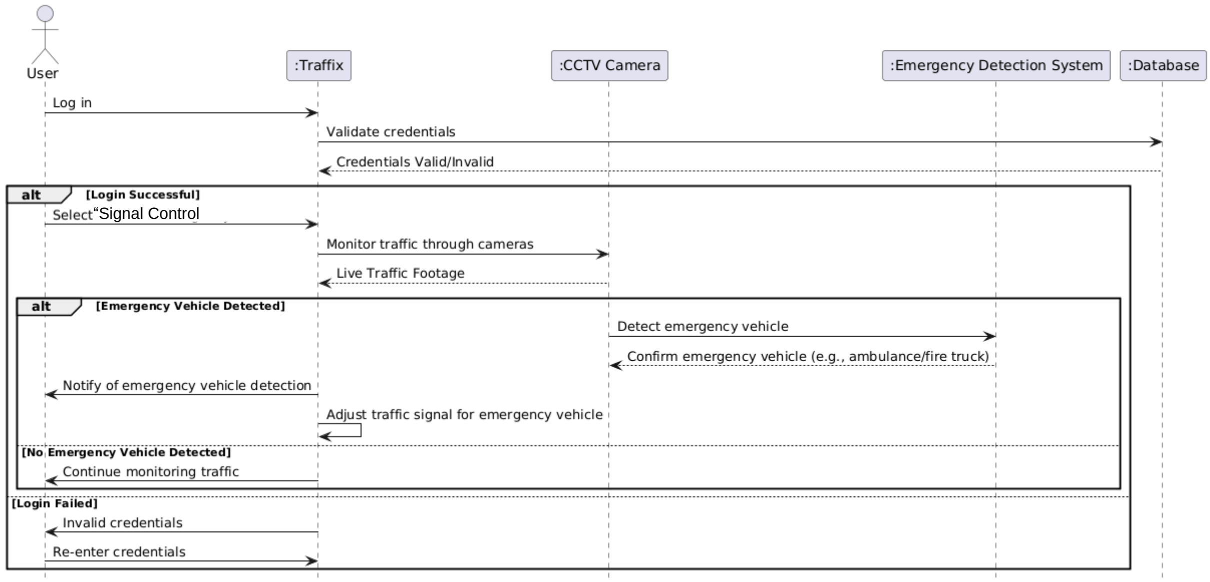



*Figure 15: Emergency Vehicle detection SD*

### 3.2.5 State Transition Diagram

In order to show how each component of the system works and responds back we have made use of state transition diagram from each state of the module.

#### Update Profile

The following is the state transition diagram for update profile.

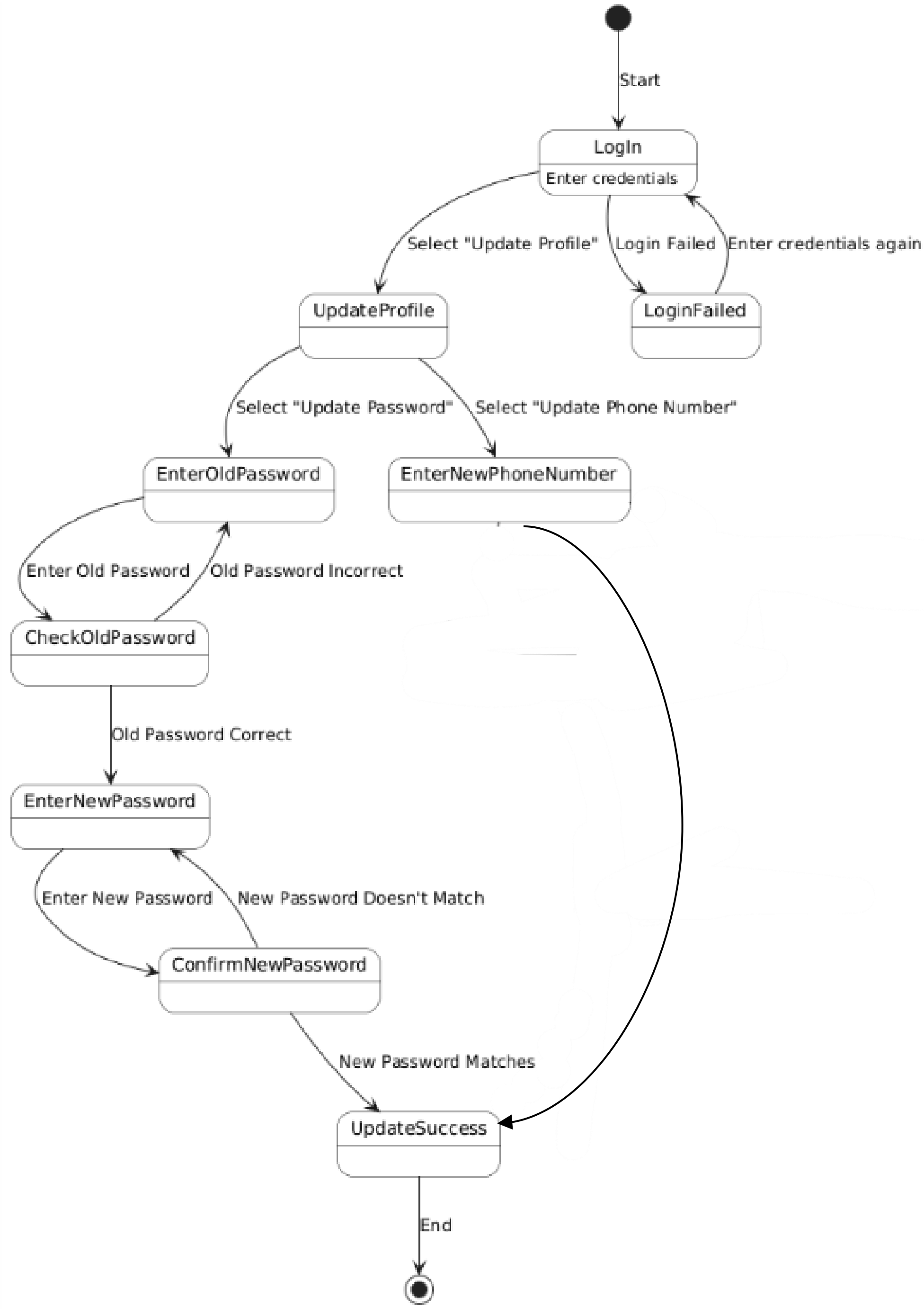



*Figure 16 Update Profile State Transition Diagram*

## Monitor Car Detection

The following is the state transition diagram for monitor traffic.

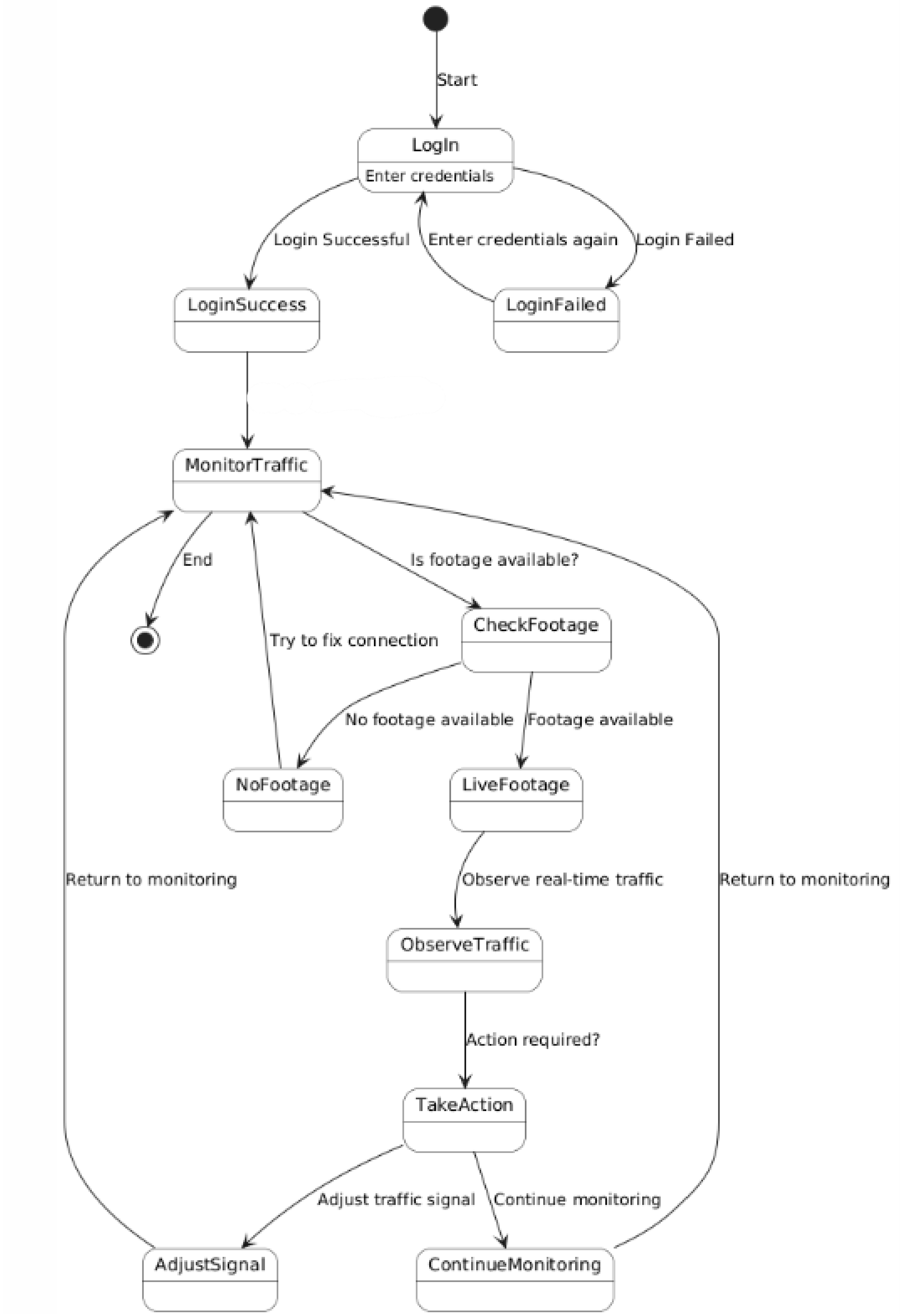



*Figure 17 Monitor Car Detection State Transition Diagram*

## Detect Emergency Vehicle

The following is the state transition diagram for detection of emergency vehicle.

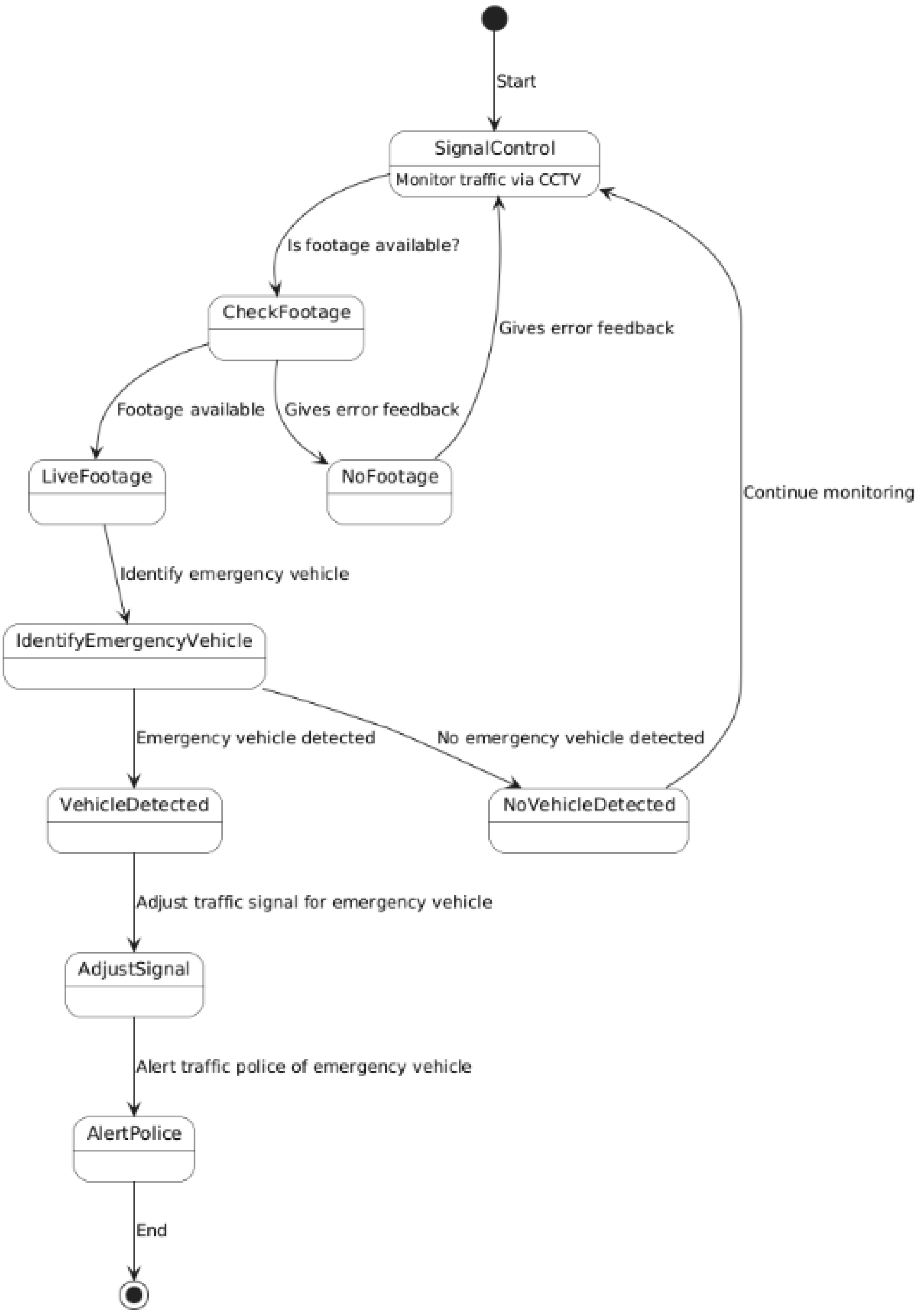



*Figure 18 Emergency Vehicle Detection State Transition Diagram*

# 3.3 Prototypes

Here are the prototypes we made for our web application before starting our project in order to gain an idea of what we will be working on and what kind of project we want to execute. We aim to create a simple UI considering that our project is complex with its technical challenges so we want to keep our UI simple, and user friendly for its targeted users I.e- Traffic Police officers, Admin.

We have included the following screens:

- Log In
- New User
- Homepage
- Signal Control (We will be showing the Monitor CCTV only on this screen for this iteration)
- How it works?

Log In
The following is the log In page prototype to authenticate user.

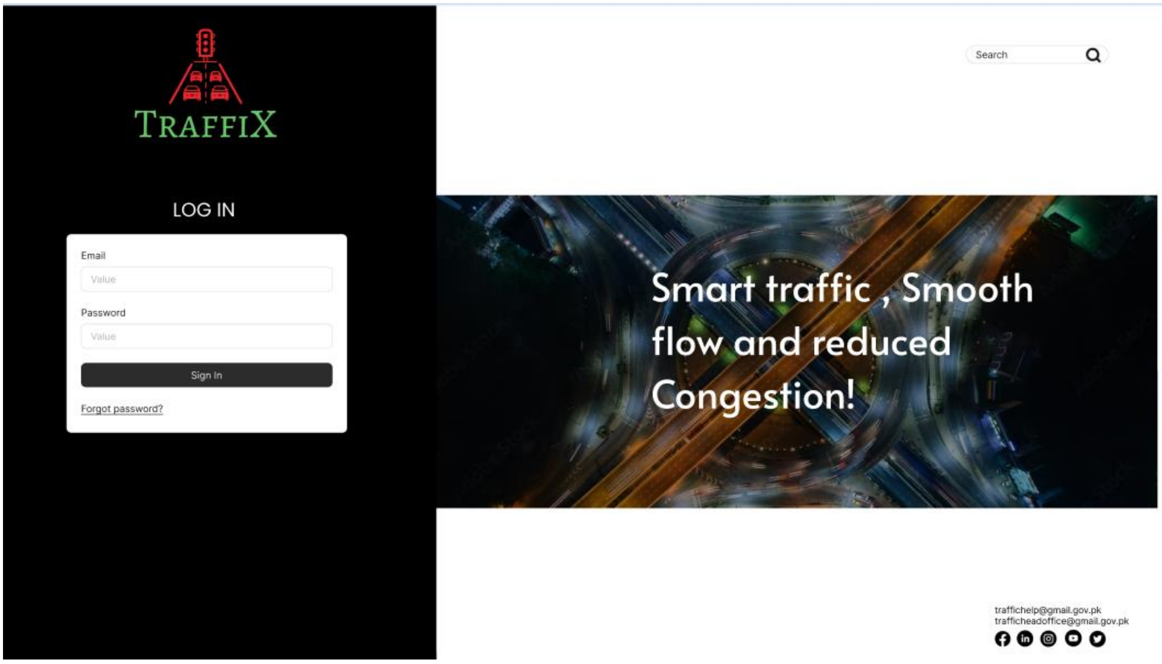



*Figure 19 Log In Prototype*

New User

The following is the register new user prototype which will only be present in the admin portal as only the admin can register new Traffic police officers and assign them location.

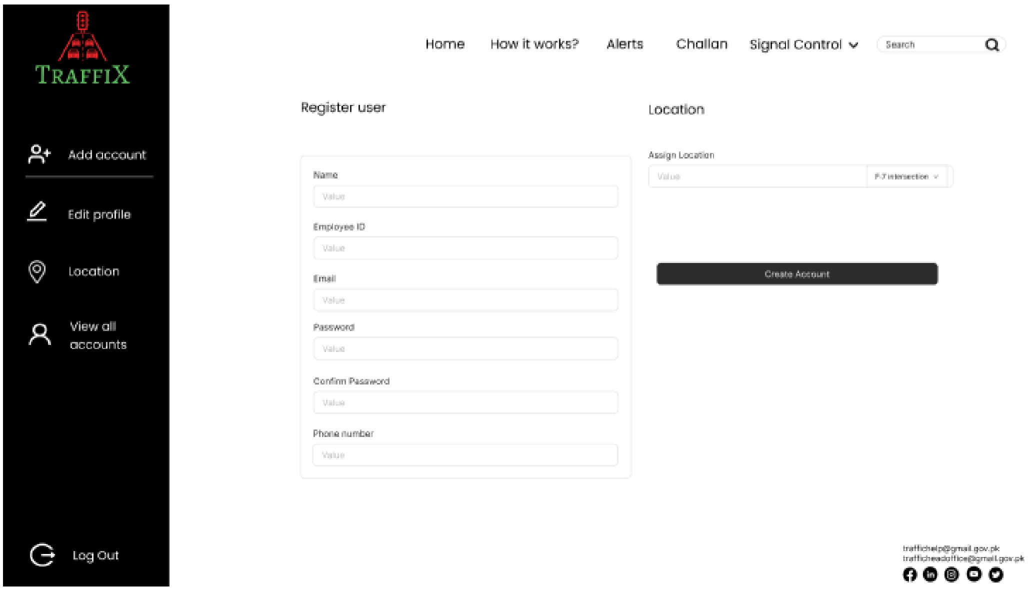



*Figure 20 New User Prototype*

Homepage

This is the homepage where user is directed to after logging in to the Traffix System.

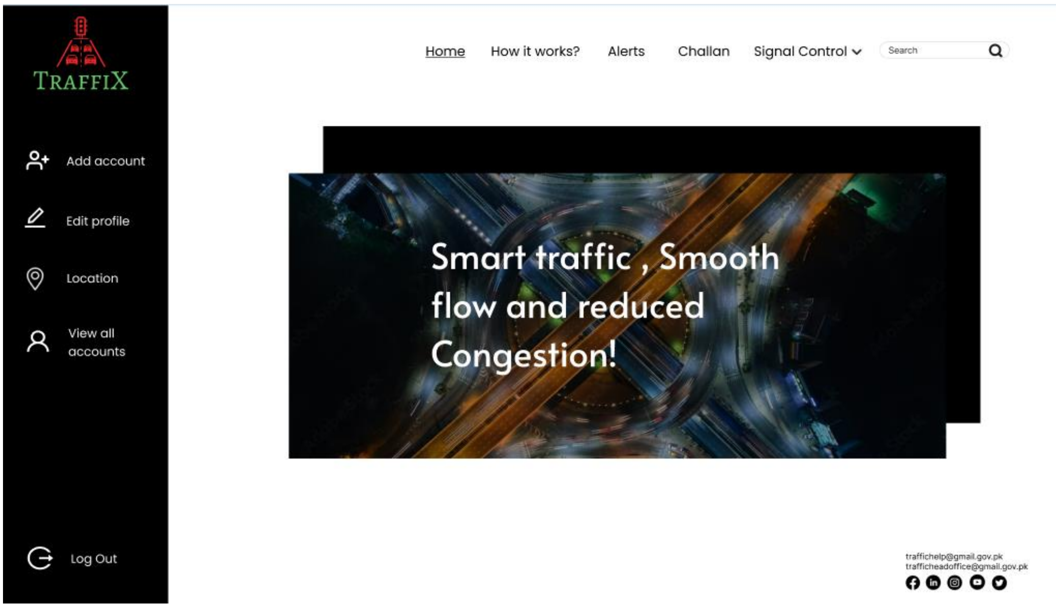



*Figure 21 Homepage Prototype*

Signal Control (Monitor CCTV footage)

This is the main screen which monitors and controls everything in Traffix. It will show the entire cycle of the traffic signal as well as the current traffic light. It will have the option to be controlled by the user or let it control automatically.

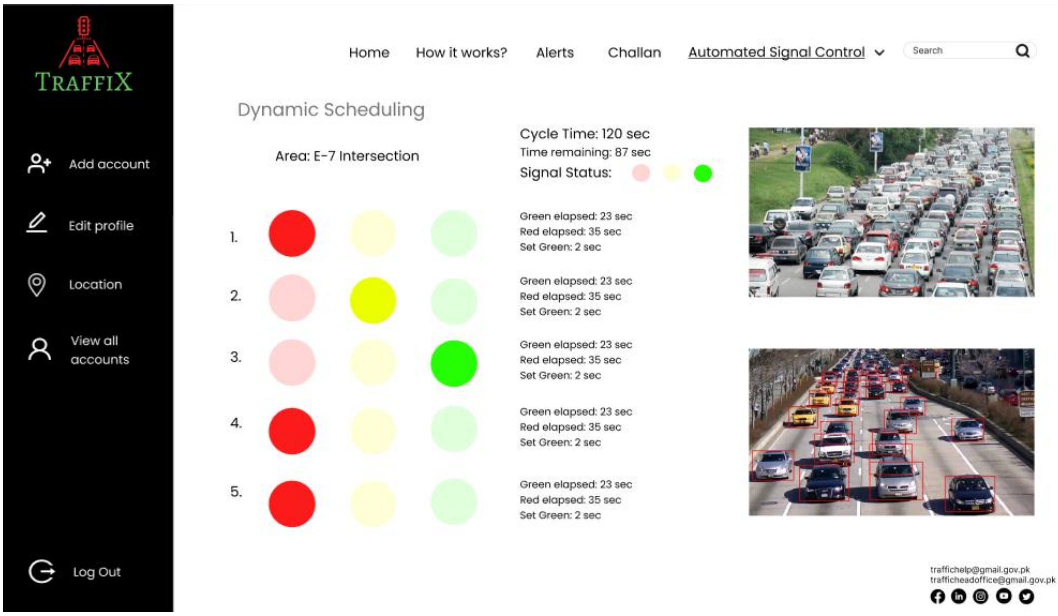



*Figure 22 Signal Control Prototype*

How it works?

This is the screen which shows how the system works for traffic police officer's ease.

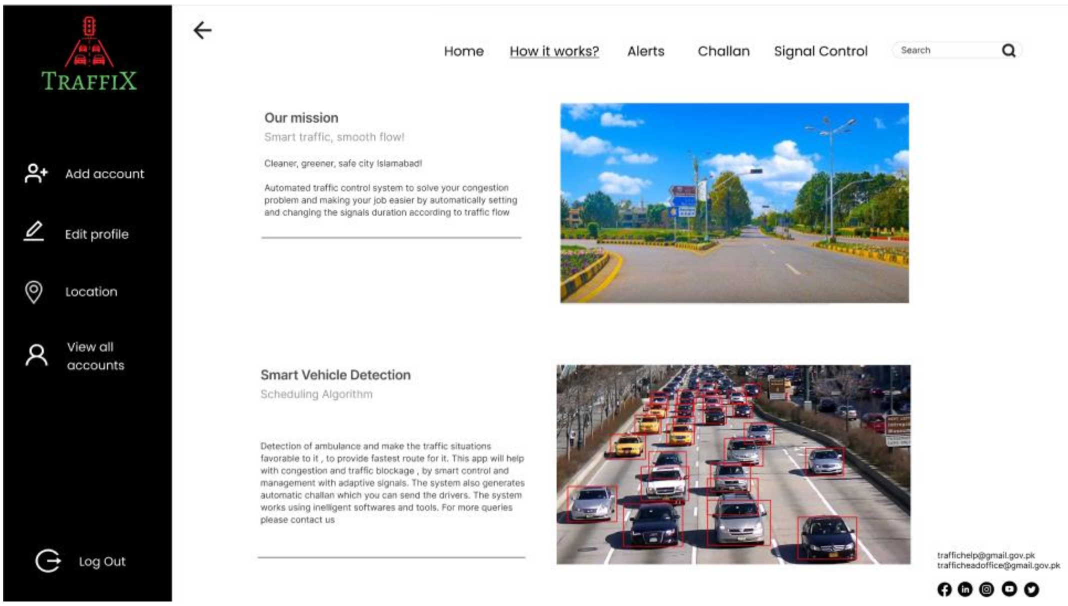



*Figure 23 How it works? Prototype*

# 3.4 Implementation

## Car Detection

Firstly here is the screenshot for car detection. We chose the yolov5 model for car detection as it has the best perfomance as compared to other versions. It is also compatible with other tools. We are running the pythorch code on our system to detect cars. We used the command ***python detect.py --weights best.pt --img 640 --conf 0.25 --source C:/Users/zaibu/Desktop/Video.mp4 --classes 2 to detect the cars.*** It is storing the data in runs\detect\exp5 file.

```
PS C:\Users\zaibu\Desktop\yolov5> python detect.py --weights yolov5s.pt --img 640 --conf 0.25 --source C:/Users/zaibu/De
sktop/Video.mp4 --classes 2
detect: weights=['yolov5s.pt'], source=C:/Users/zaibu/Desktop/Video.mp4, data=data\coco128.yaml, imgsz=[640, 640], conf_
thres=0.25, iou_thres=0.45, max_det=1000, device=, view_img=False, save_txt=False, save_format=0, save_csv=False, save_c
onf=False, save_crop=False, nosave=False, classes=[2], agnostic_nms=False, augment=False, visualize=False, update=False,
 project=runs\detect, name=exp, exist_ok=False, line_thickness=3, hide_labels=False, hide_conf=False, half=False, dnn=Fa
lse, vid_stride=1
YOLOv5  v7.0-368-gb163ff8d Python-3.11.9 torch-2.4.1+cpu CPU

Fusing layers...
YOLOv5s summary: 213 layers, 7225885 parameters, 0 gradients, 16.4 GFLOPs
video 1/1 (1/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 7 cars, 292.7ms
video 1/1 (2/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 7 cars, 262.8ms
video 1/1 (3/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 7 cars, 216.8ms
video 1/1 (4/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 7 cars, 285.6ms
video 1/1 (5/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 7 cars, 281.2ms
video 1/1 (6/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 6 cars, 262.8ms
video 1/1 (7/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 6 cars, 272.4ms
video 1/1 (8/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 6 cars, 245.0ms
video 1/1 (9/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 6 cars, 242.3ms
video 1/1 (10/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 6 cars, 251.9ms
video 1/1 (11/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 6 cars, 222.8ms
video 1/1 (12/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 6 cars, 275.1ms
video 1/1 (13/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 5 cars, 227.7ms
video 1/1 (14/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 5 cars, 258.8ms
video 1/1 (15/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 7 cars, 229.9ms
video 1/1 (16/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 8 cars, 240.6ms
video 1/1 (17/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 9 cars, 235.4ms
video 1/1 (18/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 9 cars, 265.0ms
```

```
Windows PowerShell
video 1/1 (424/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 4 cars, 550.3ms
video 1/1 (425/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 3 cars, 761.5ms
video 1/1 (426/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 3 cars, 735.4ms
video 1/1 (427/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 2 cars, 654.0ms
video 1/1 (428/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 3 cars, 605.8ms
video 1/1 (429/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 2 cars, 661.6ms
video 1/1 (430/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 2 cars, 615.2ms
video 1/1 (431/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 2 cars, 852.6ms
video 1/1 (432/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 3 cars, 325.3ms
video 1/1 (433/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 3 cars, 704.4ms
video 1/1 (434/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 3 cars, 774.1ms
video 1/1 (435/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 3 cars, 735.0ms
video 1/1 (436/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 3 cars, 734.3ms
video 1/1 (437/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 3 cars, 724.0ms
video 1/1 (438/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 4 cars, 574.7ms
video 1/1 (439/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 2 cars, 680.1ms
video 1/1 (440/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 2 cars, 875.3ms
video 1/1 (441/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 2 cars, 521.8ms
video 1/1 (442/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 3 cars, 870.7ms
video 1/1 (443/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 3 cars, 447.5ms
video 1/1 (444/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 3 cars, 814.1ms
video 1/1 (445/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 2 cars, 560.8ms
video 1/1 (446/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 3 cars, 710.7ms
video 1/1 (447/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 4 cars, 662.9ms
video 1/1 (448/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 3 cars, 716.4ms
video 1/1 (449/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 3 cars, 606.3ms
video 1/1 (450/450) C:\Users\zaibu\Desktop\Video.mp4: 384x640 3 cars, 436.3ms
Speed: 2.9ms pre-process, 457.3ms inference, 3.2ms NMS per image at shape (1, 3, 640, 640)
Results saved to runs\detect\exp5
PS C:\Users\zaibu\Desktop\yolov5>
```

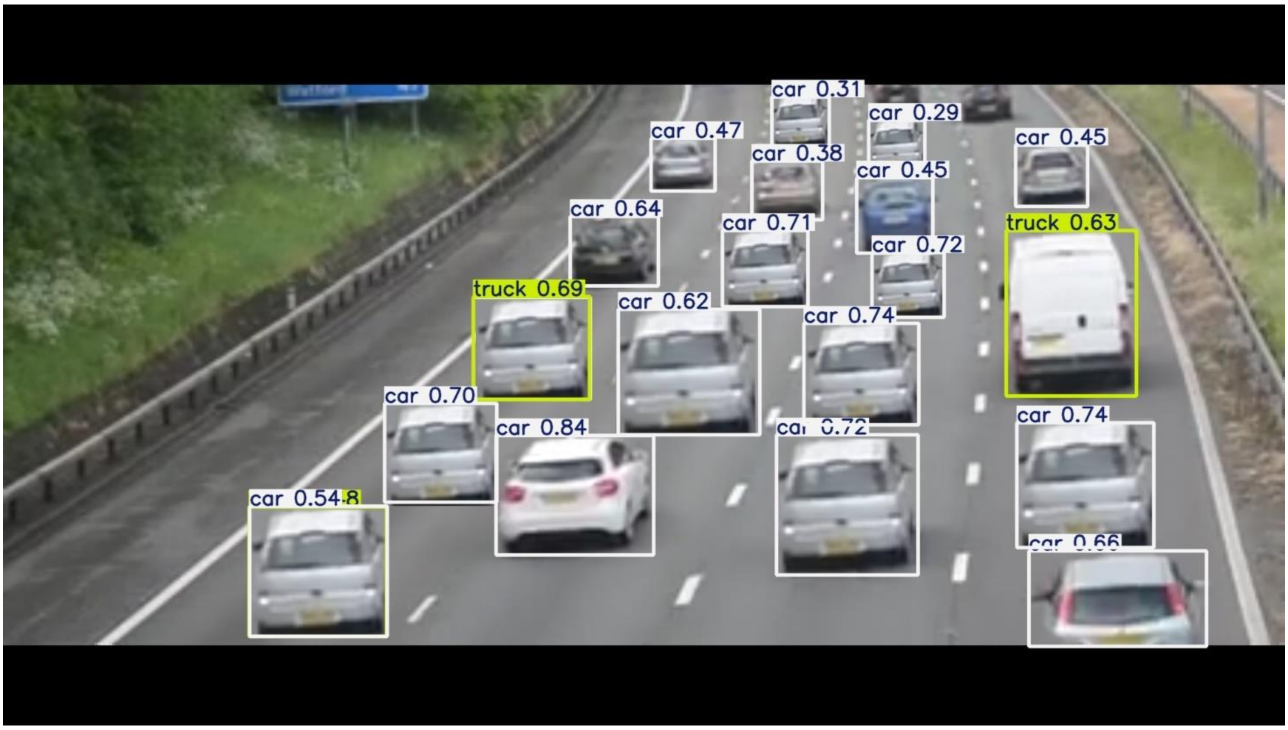
The results of the video are as follows:

**Before**


**After**: It is identifying the cars and their accuracy.


It also has the option to detect cars in *live* camera however to test it out we are using the database we gathered in this iteration. Below are the results:

*Figure 24 Car detection- Before & After*


car 0.31
car 0.29
car 0.47
car 0.38
car 0.45
car 0.45
car 0.64
car 0.71
truck 0.63
car 0.72
truck 0.69
car 0.62
car 0.74
car 0.70
car 0.84
car 0.74
car 0.54


car 0.89
car 0.81


21.21320
View Menu
car 0.92


KEEP LEFT UNLESS OVERTAKING
car 0.31
car 0.36
truck 0.349
car 0.82
car 0.88
car 0.72
car 0.76

Emergency Vehicle detection

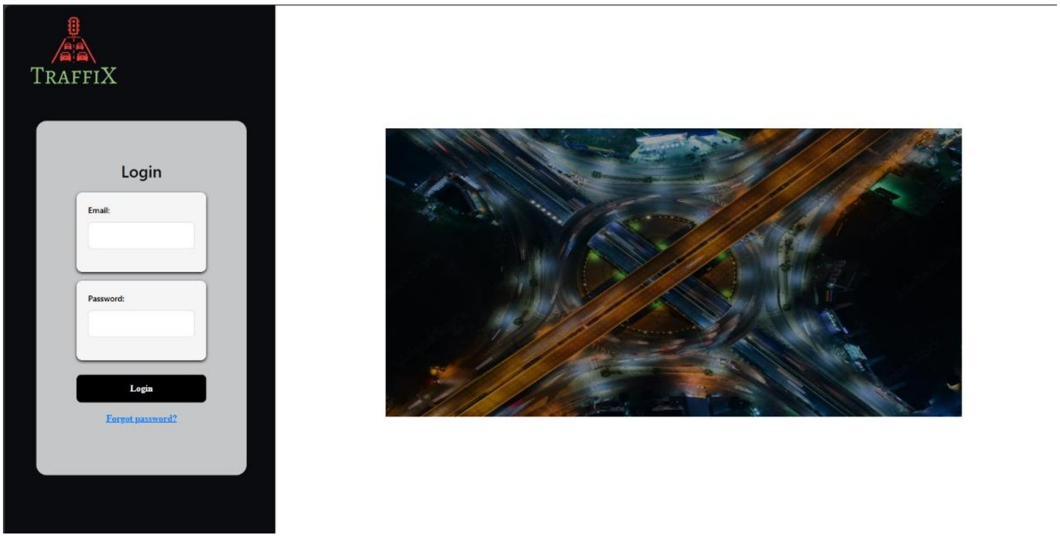
**For emergency vehicle detection we fine-tuned the same pre-trained model with emergency vehicle images that we gathered while collecting our database. Here are the results:**



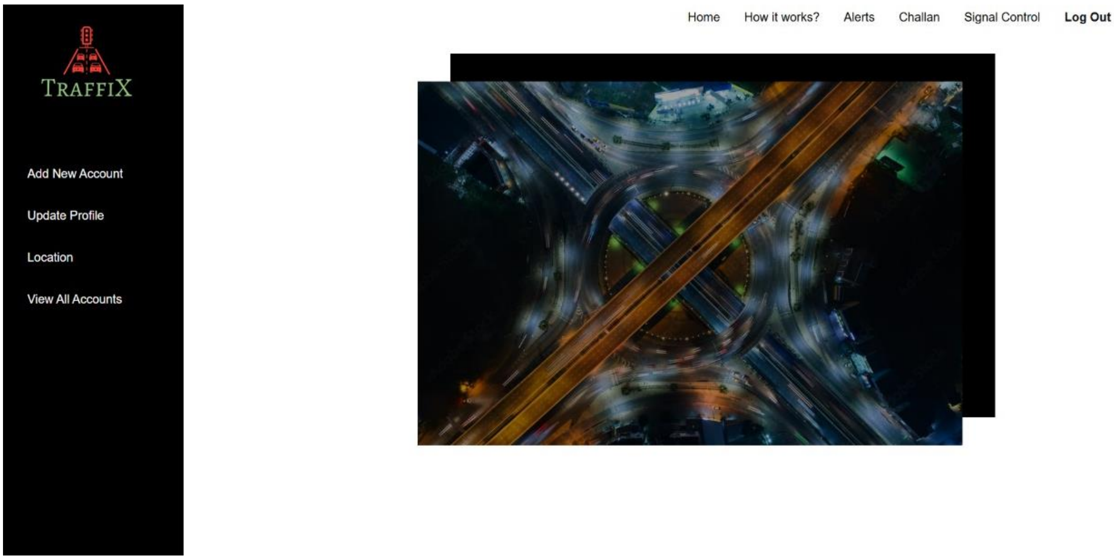
*Figure 25 Emergency Vehicle Detection*

Web Application

Here is the log in page of our system. Only pre-registered users with the email ending with “police.gov.pk” can log in here to access the system.

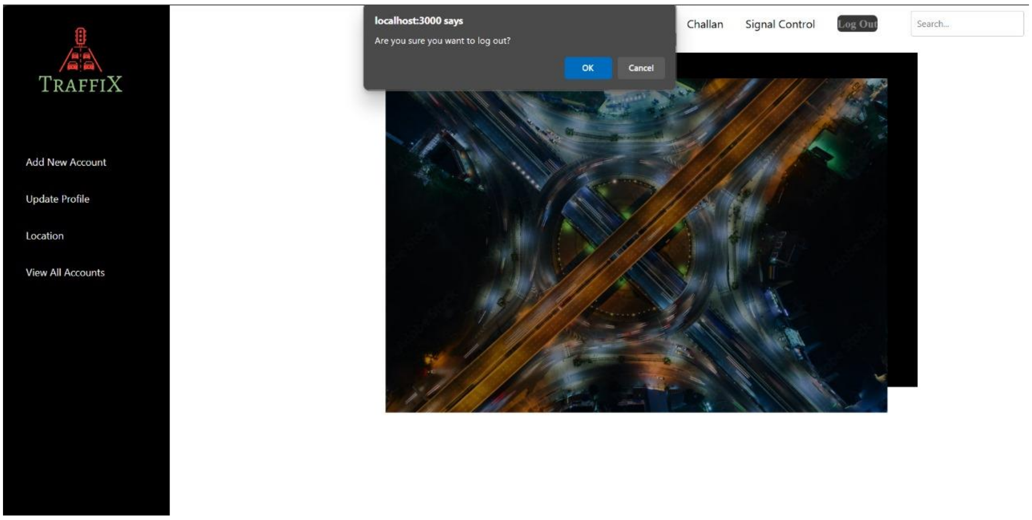



This is the homepage where user is directed to after logging in to the Traffix System.

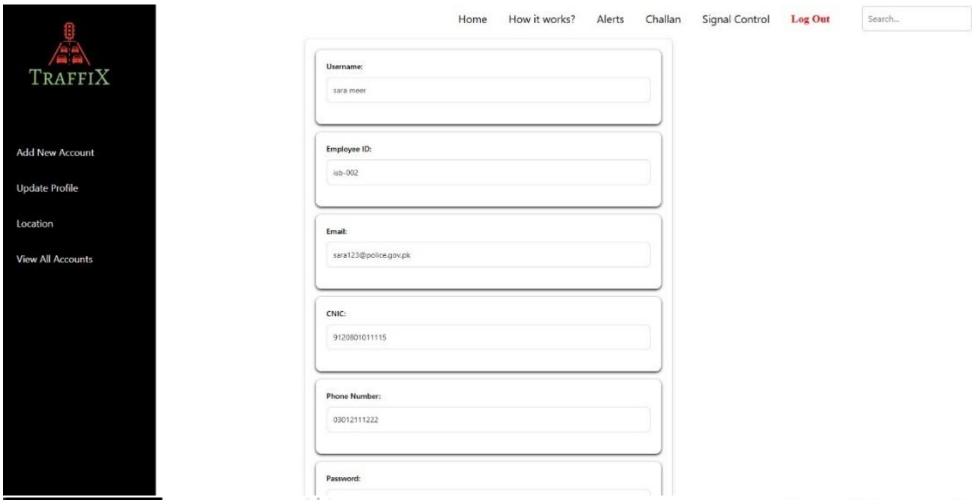


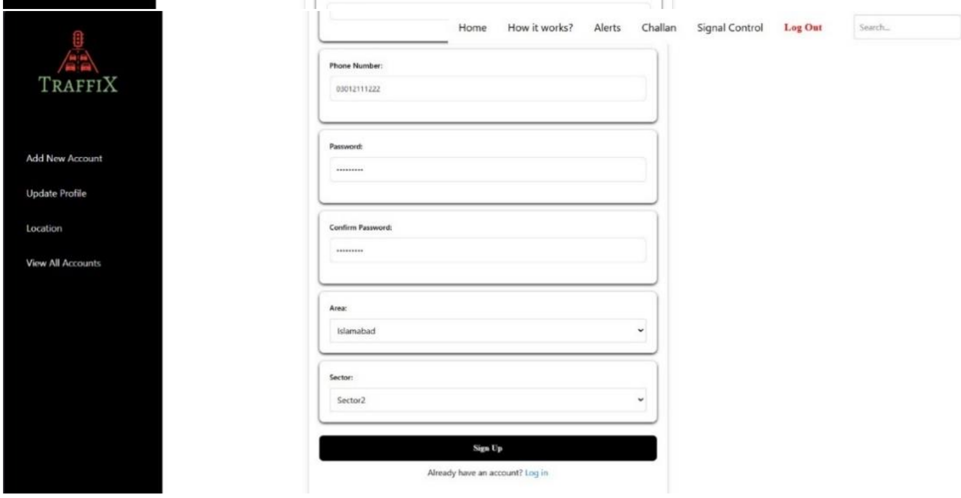

It is the central point for all the options for the user who has just logged in to the system. The user can access all the features including how it works, alerts, challan, signal control, log out etc from the navigation bar in top and side bar on the left. On the left side all the features will be presented i.e- register new user, update your own profile, View all accounts etc.

Upon clicking on the log out button it prompts the user to re-confirm their action.

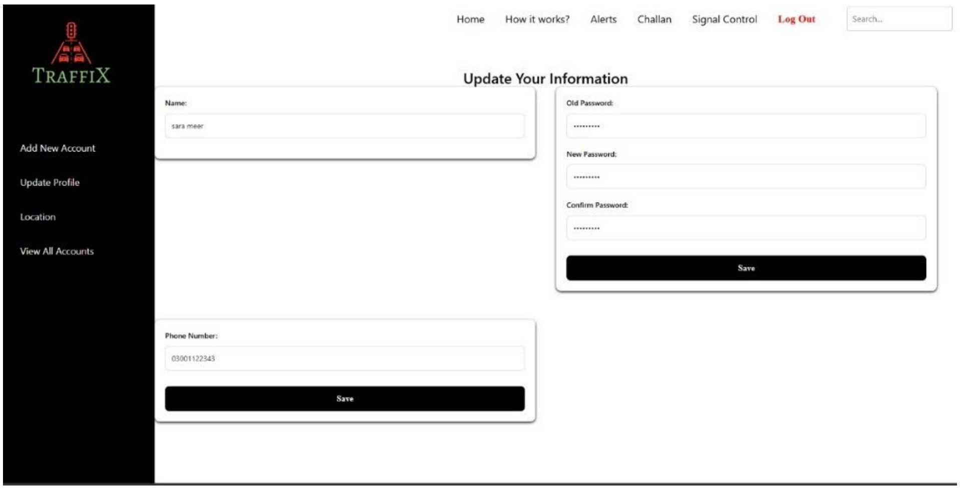



This is the register new page which will only be accessed by the admin. It has various checks, and takes input to register the new Traffic Police officer.



Th update your profile page allows you to view your name and update you phone number and password to keep your system safe and secure.



# 3.5 Data Design

**Data Storage and Processing:**

The data is stored in  Mongodb Atlas database in which we are doing the data storage, retrieval, updation, and deletion.

The data will be organized into tables based on the entities and their attributes.

**Database: Mongodb Atlas**

Tables: Admin, Officers, TrafficSignals, , Challan, Vehicle.

Columns of the tables: Attributes of each entity (eg username, password, email, CNIC, city, etc.)

- **Admin table:** has the information about the admin, i.e. username, password, email, CNIC, city, and sector.
- **Officers table:** has the information about traffic officers, i.e. username, password, email, CNIC, city, and sector.
- **TrafficSignals table:** has the information about traffic signals, i.e. signalID, signalStatus (red, yellow, green), timing (time left of the signal), and location.
- **Challan table:** has the information about traffic violation tickets for crossing the boundary line, i.e. challanID, challanNumber, challanAmount, issueDate, and challanStatus (if it has been paid or if it is still pending).
- **Vehicle table:** has the information about vehicles on the roads for the traffic signal violation, i.e. vehicleID, vehicleType (to check if it is a normal car or an emergency vehicle), vehicleNumber (incase of challan).

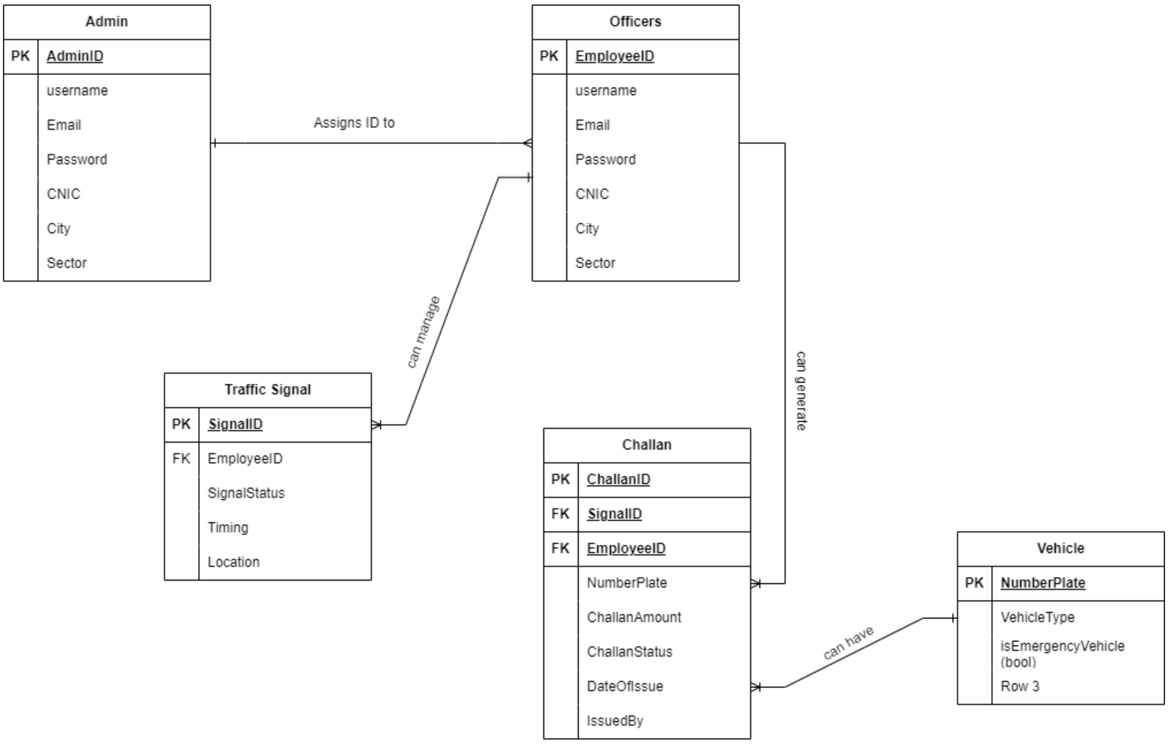



*Figure 26 Data Design -ERD*

# 3.6 Iteration Timeline

Our project is divided into four iterations each of three months, starting from August. All the work is equally distributed amongst the group members and will be done collaboratively. Our focus for this iteration was iteration 1.

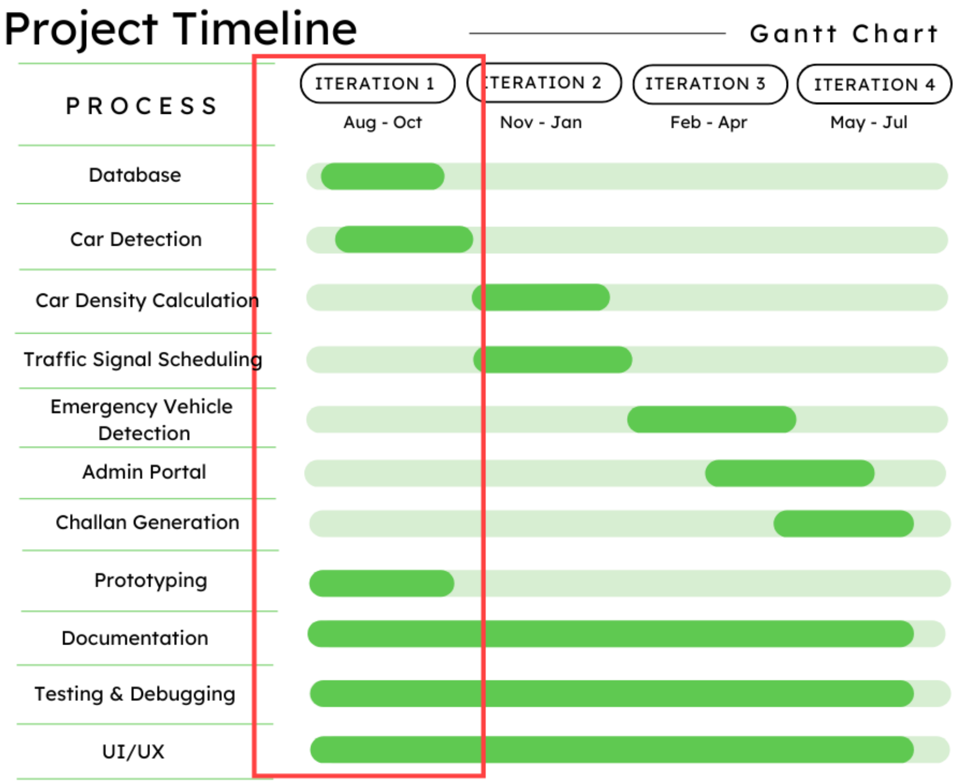



*Figure 27 Work Timeline*

# 3.6 Conclusion

In conclusion, we implemented the iteration 1 update profile, monitor traffic using vehicle detection including emergency vehicle detection to identify the type of vehicle. However, we will be adjusting the traffic signal for the emergency vehicle in iteration 3 to prioritize traffic lane. For the detection and identification of vehicle we used YOLO algorithm, fine-tuned and trained our model with images/videos for Emergency vehicle detection. We have successfully implemented all the features of iteration one including database collection, prototyping, UI/UX and car detection and have a working demo of our system, with all the features mentioned above.

# Bibliography

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